

Power generation scenarios for Cameroon: Valorisation of biomass for the reduction of electricity transmission and the mitigation of greenhouse gas emissions by 2050

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ABSTRACT

This work aimed at solving the environmental problem of the hottest greenhouse gas reduction. The paper identifies the energy electricity demand, proposes solutions, and calculates the greenhouse gases emitted in each of the interconnected networks available in Cameroon. For a total production estimated at 1292 MW, the country's electricity production depends heavily on hydropower plants (57%), thermal power plants (42%), and only 1% of biomass. Specifically, biomass would contribute to 40% of the supply of the demand if it is sufficiently exploited. Hydroelectricity needs transmission for long distances. The electricity demand in the median economic scenario is the basis of this study. GHGs were calculated using the tool MESSAGE. The results of this study show that it will be possible, without taking into account the fact that biomass is clean energy, to avoid 462 kt of CO₂eq and 14 kt of CO₂eq for South Interconnected Network and East Interconnected Network respectively. The amounts of CH₄, SO₂, and NO_x are not statistically significant. This study takes into account the main existing energy sources in the country. The specificity of this study is to treat the three interconnected networks of the country separately. The use of biomass in local power plants was proposed to reduce dependence on fossil fuels. This study shows that to reduce the GHG emission and the loss of electricity due to its transmission in the country, it is better to use biomass instead of fossil fuels for the production of electricity in Cameroon.

1. Introduction

The Cameroon population is increasing. It increases every year by about 2.6% (INS, 2019). A study by the National Institute of Statistics shows that in 2014, Cameroon's population was 21 million. In 2035, it is projected to be around 31 million. This demographic growth will cause an increase in electricity demand. Electricity has become necessary for daily life and is an essential factor for company competitiveness. However, the electricity sector in Cameroon is characterized by an imbalance between supply and demand. Electricity production is still heavily dependent on petroleum products (heavy fuel oil and light fuel oil), which produce a lot of greenhouse gases. Since 2018, the current ratio of power generation from different energy sources is, 57% for the hydropower plants and 43% for the fossil fuels (21% gas and 22% LFO and HFO) Table 1 (SIE, 2018). The CO₂, CH₄, and NO₂ emissions from fossil fuels used and other sources in the country are shown in Table 2. This later table shows that the combustion of fossil fuels is the main source of GHG emissions, followed by the transport sector. This means, that the

development and use of biomass energy can promote and develop a country's economy and enhance environmental protection (Guozhu et al., 2018).

Africa is facing the challenge of the energy transition, which requires changes in its consumption patterns to address two distinct issues: the loss of its organic capital (mainly forestry) and its extreme dependence on hydrocarbons or coal, which for most countries on the continent have to be imported.

In African countries, this transition has two components: one concerns the part of the population that uses biomass massively for energy (mainly cooking) and consumes little or no fossil fuels; the other concerns the population that consumes a significant amount of fossil fuels in the form of gas, electricity or petroleum fuel, and therefore uses little biomass.

In the Central African sub-region, bioenergy has a very specific position. On the one hand, the public authorities are paying little attention to wood energy, charcoal, biogas, the recycling of urban waste (into compost, for example), or solar energy. This applies to the whole of the region, and in particular to North Cameroon, which is experiencing the

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Nomenclature		λ_{lm}	Length of load region lm , the load region with maximum capacity requirements, as a fraction of the year
i_{yr}	Base year (yr)	Abbreviations	
ν	Length of the periods in that the time before the base year is divided (yr)	GHG	Green House Gas
τ	Plant life in years (yr)	MESSAGE	Model for Energy Supply Strategy Alternatives and their Environmental impact
c_o	Total capacity in the base year (MW)	CO ₂ eq	Carbon Dioxide equivalent
f_i	Capacity variable is continuous, and represents the minimum installed capacity per year (unit size) if the variable is an integer (MW)	CH ₄	Methane
γ	Annual growth of new installations before the base year (%)	SO ₂	Sulfur Dioxide
y_t	Annual construction in the period t (yr)	NO _x	Nitrous oxide
ν	Number of years the technology exists after the end of the calculation horizon	INS	National Institute of Statistics
dr_τ	Discount rate for year τ (%)	LFO	Light Fuel Oil (Gas oil)
τ_p	Plant life in years	HFO	Heavy Fuel Oil
C_t	Investment cost in year t (US\$'00/kW)	NO ₂	Nitrogen Dioxide
C'_t	Reduced investment (%)	CO ₂	Carbon Dioxide
Q_r	Reference demand level	PANERP	Plan national de l'énergie pour réduire la pauvreté
P_r	Reference price level (US\$'00/kW)	SOSUCAM	Société Sucrière du Cameroun
ϵ	Elasticity, (assumed to be < 0)	SOCAPALM	Société Camerounaise de Palmeraie
z_{svd}	Activity of conversion technology ν in period t and, if defined so, load region l	PDSE2030	Plan de Developpement du Secteur de l'Energie à l'horizon 2030
$Y_{zsvd..t}$	Capacity variable of conversion technology ν (MW)	RIS	Réseau Interconnecté SUD
ϵ_{svd}	Efficiency of technology ν in converting the main energy input, s , into the main energy output, d , (%)	RIN	Réseau Interconnecté NORD
k_{svd}	Last period in that technology ν can be constructed (yr)	RIE	Réseau Interconnecté EST
π_{svd}	Plant factor of technology ν , having different meaning depending on the type of capacity equation applied, (%)	IAEA	International Atomic Energy Agency
$\Delta\tau$	Length of period τ in years	PDSE	Plan de Developpement du Secteur de l'Energie
τ_{svd}	Plant life of technology ν in periods (yr)	IPPC	Integrated Pollution Prevention and Control
hc_{svd}^d	Installations built before the time horizon under consideration that are still in operation in the first year of period t	TWh	Térawattheure
l_m	Load region with maximum capacity use if the production pattern over the year is fixed	MINEE	Ministère de l'Eau et de l'Energie
$\pi(l_m,svd)$	Share of output in the load region with maximum production (%)	GW	Gigawatt
$re_{\sigma\delta}^{Fsvd}$	Relative capacity of the main output of technology (or operation mode) svd to the capacity of the main output of the alternative technology (or operation mode) $\sigma\delta$ (MWyr)	ENEO	Energy of Cameroon
λ_l	Length of load region l as a fraction of the year	EIA	Energy Information Administration
		IMF	International Monetary Fund
		SIE	Situation Energétique du Cameroun
		IIASA	International Institute for Applied Systems Analysis
		INS	Institut National de Statistiques
		Mxg	matrix generation program
		opts	Optimization program
		tdb	Technology database
		adb	Application database
		upd	Update database
		LCA	Life Cycle Analysis
		COP 20	Conferences of the Parties

difficulties of access to energy more acutely than other regions because of its soil and climate characteristics, which are similar to those of the Sahel. On the other hand, agrofuels (mainly from palm oil or jatropha) are part of a non-transparent process of land acquisition, often on a large scale.

Overall, more than 65% of Cameroon's energy consumption is met by traditional energy sources (wood, coal, etc.), 21% by petroleum products, and 14% by electricity, according to the PANERP (Gabas, 2014). Nearly all poor households in Cameroon use firewood to meet their energy needs for cooking. Two sources of renewable energy are used, meeting more than 80% of overall energy needs: biomass (mainly wood and coal) accounts for two-thirds of these renewable energies, and hydroelectricity for a third. Solar energy is used very little. In the far north of Cameroon, household energy supplies depend on forest biomass for more than 95% of their needs, and the vast majority of these are met by the informal sector (Gabas, 2014). Some industries produce electricity from their waste. These include SOSUCAM, SOCAPALM.

Therefore, the country must use other sources of energy (bioenergy), such as biomass. The renewable energy sector is still underdeveloped in Cameroon. Wood energy is generally used in households. 82.6% of households use wood, charcoal, sawdust, or chips as their main cooking energy. Cameroonians use rudimentary methods (direct combustion, in traditional open fireplaces) to extract the energy contained in biomass. However, these methods are less efficient. The biomass resources in Cameroon have high potential. Sufficiently exploited, biomass could contribute to 40% of Cameroon's electricity demand (Ackom et al., 2013). Thus, there is a need to diversify the sources and technologies of electricity generation in Cameroon. Biomass can be an unstable source compared to fuel oils and is proposed as a backup solution. Indeed, energy prices often increase, and sometimes supplies are disrupted, and a large part of this fuel comes from outside. The use of biomass can affect food stability as it can be designed in some cases. The study therefore only valorizes waste.

Biomass, mainly used in the residential sector for heat production,

Table 1
Energy balance of Cameroon.

Energy balance (2018)	
Primary energy supply (TPES)	7,6 M tep (318,3 PJ)
By energy source	Oil: 22,5% Natural gas: 7,3% Electricity: 5,7%
Renewable energy	5,7%
Total consumption:	6,5 M tep (273,2 PJ)
By inhabitant	0,3 tep/hab (12 GJ/hab)
By sector	Households: 69,1% Industry: 6% Transport: 16,5% Services: 8,3% Agriculture: 0,1% Fishing: 0%
Electricity (2018)	
Production	6,92 TWh
By sector	Hydro: 73,2% thermal: 25,7% Biomass/ waste: 1%

Table 2
Energy-related GHG emissions in 2018.

Designation	CO ₂ (Gg)	CH ₄ (Gg)	NO ₂ (Gg)	TOTAL (Gg CO ₂ eq.)
Combustion of fuels	3 983,0	141,8	2,0	7 584,4
Energy Industries	1 383,8	0,9	0,1	1 438,9
Manufacturing and construction industries	71,0	0,0	0,0	71,0
Transportation	2 163,9	0,0	0,0	2 169,3
Civil Aviation	39,4	0,0	0,0	39,4
Road transport	2 058,4	0,0	0,0	2 063,9
Rail transport	46,7	0,0	0,0	46,7
Inland navigation	19,4	0,0	0,0	19,4
Other sectors	364,2	141,0	1,9	3 905,2
Commercial and Institutional	10,1	12,6	0,2	327,4
Residential	354,1	128,4	1,7	3 577,7
Agriculture/Forestry/Fishing	0,0	0,0	0,0	0,0
Fugitive Emissions	0,0	0,7	0,0	14,5
Oil & Gas	0,0	0,7	0,0	14,5
TOTAL EMISSIONS	3 983,0	142,5	2,0	7 598,9

dominates the energy mix of Cameroon. However, its usage for electricity production remains marginal (PDSE2030, 2015). Whereas, Cameroon's biomass potential is enormous, resulting from domestic waste, forest products, and their waste, and from agricultural waste. Different biomass fuels are utilized for bioenergy power generation, such as agricultural residues, wood chips, wood residues, specifically grown energy crops (e.g., miscanthus, switchgrass), as well as waste materials (Alice et al., 2018).

Indeed, it is necessary to introduce biomass into the energy mix of the country because it is clean and renewable energy. It is a very common source of energy in the country, but its use for electricity generation remains low. The waste in the environment often causes diseases that have negative effects on human health. In Cameroon, biomass is available in the form of biomass with high and low moisture content. Biomass with high moisture content combines waste from households, markets, and food processing industries. Biomass with low moisture content comes from forestry operations (wood processing industries, forest residues). Another advantage of this resource is that its recovery will allow the cleaning of our cities which are sinking under the weight of waste. The quantity and quality of biomass with low moisture content for power generation needs are shown in Table 3, Table 4, and Table 5. The introduction of this source will finally reduce dependence on fossil fuels and reduce GHG emissions. The Thermal oil power plants can be gradually replaced by biomass power plants. This will have the main impact on reducing GHG emissions. This work thus proposes a plan for the expansion of biomass power plants in the RIS and RIE. However, the choice of technologies depends on several factors.

Technologies are important factors in the power generation system. The different power generation technologies have very different economic costs, run times, and emission factors. The study by Alice et al. (2018) highlights the feasibility of biomass power plants. This biomass is transformed in the following ways: anaerobic digesters, combustors, gasifiers, pyrolyses etc. Therefore, the choice of technologies and their role in an energy system is critical. It must be based on cost, energy efficiency, and environmental effects on the overall energy system. Technology selection is a comprehensive decision-making process. It depends on several factors, such as technology cost, demand, resource constraints, and environmental standards. Many policies and regulations have been formulated by the government. This is to reduce the cost of technology implementation directly and indirectly. Also, choose the ones which are low consumption and low carbon emission. Each technology or plant has a well-defined GHG emission factor for this purpose.

The search for energy has led to greenhouse gas (GHG) emissions that have contributed significantly to problems such as global warming.

Table 3
Estimated energy potential from agricultural production, Western region (Source EED).

Culture	Annual production (t)	Department	Annual production (t)	Available waste quantity (t)	Available primary energy (Kcal)	Available annual electricity potential (GWh)
Corn	79566	Banbouts/Haut-Nkam	19433	19433	6.8 * 1010	23.70
		Haut-plateaux/Koung-Khi	11678	11678	4.1 * 1010	14.30
		Menoua	32223	32223	1.1 * 1011	38.37
		Mifi/Nde	5278	5278	1.8 * 1010	6.23
		Noun	10954	10954	3.8 * 1010	13.25
Rice	1101	Ndé	718	179	5.1 * 108	0.19
		Noun	383	95	2.8 * 108	0.09
Peanut	4 062	Banbouts/Haut-Nkam	916	183	4.6 * 108	0.16
		Koung-Khi	601	120	3 * 108	0.10
		Haut-plateaux	1803	360	9 * 108	0.31
		Menoua	742	148	3.7 * 108	0.13
Oil palm	4907	Haut-Nkam	1112	467	4.9 * 108	0.17
		Menoua	2029	852	8.9 * 108	0.31
		Ndé	655	275	2.9 * 108	0.10
		Noun	1111	467	4.9 * 108	0.17
Annual electricity potential available for the WEST zone						95.58

Table 4
estimated energy potential from agricultural production, Littoral region.

Culture	Annual production (t)	Department	Annual production (t)	Available waste quantity (t)	Available primary energy (Kcal)	Available annual electricity potential (GWh)
Corn	86833	Moungo	56 205	56 205	$2.0 * 10^{11}$	69.76
		Nkam	17 951	17 951	$6.3 * 10^{10}$	21.98
		Sanaga maritime	7 953	7 953	$2.8 * 10^{10}$	9.76
		Wouri	4 724	4 724	$1.6 * 10^{10}$	5.58
Peanut	2 044	Moungo	543	108	$2.7 * 10^8$	0.09
		Nkam	601	120	$3.0 * 10^8$	0.10
		Sanaga maritime	812	162	$4.0 * 10^6$	0.14
		Wouri	88	18	$4.5 * 10^7$	0.01
Oil palm	36 147	Moungo	7 997	3 587	$3.8 * 10^9$	0.132
		Nkam	4 948	2 078	$2.2 * 10^9$	0.75
		Sanaga maritime	23 014	9 665	$1.0 * 10^{10}$	3.49
		Wouri	188	79	$8.3 * 10^7$	0.03
Annual electricity potential available for the LITTORAL area						113.02

Table 5
Forest and sawmill residues in Cameroon in 2020 (GIZ, 2020).

Annual available standing timber volume	5.000.000 m ³
Parkland residue (about 14% of the standing wood volume)	700.000 m ³
Mill input	2.500.000 m ³
Wood produced	748.000 m ³
Material yield	30%
Sawmill residue (sawdust, slabs, shavings, sapwood, heart defects.)	1.752.000 m ³ (70% of waste in sawmills)
Parkland and sawmill residues	2.45 millions m ³

Emissions of GHGs, such as carbon dioxide, methane, and others, have increased dramatically over the past century due to fossil fuel combustion and land-use changes. Bioenergy has significant greenhouse gas (GHG) mitigation power. But resources must be developed sustainably and efficient bioenergy systems must be used. Using biomass instead of conventional fossil fuels reduces CO and other pollutant emissions, making biomass-based systems a more environmentally friendly concept

(Zhenjuan et al., 2021).

In this paper, we select some key new energy generation technologies from three existing networks in Cameroon. We use the energy technology model to analyze their contributions and impacts on GHG mitigation. The difference between this work and the previous one is that in this study, the three distribution networks are considered separately. It is an effective situation in Cameroon. However, [Guemene Dountio et al. \(2016\)](#) made a global study considering that the three networks are interconnected. The Southern Interconnected Network (RIS), the Northern Interconnected Network (RIN), and the Eastern Interconnected Network (RIE). However, this study does consider that the networks are not connected. An environmental impact assessment of some power plants that were not taken into account in the previous study is also performed. The study also elaborates on the scenarios which constitute the basis of the model to be produced. Although the country has three development scenarios, this study is based only on the median scenario. To highlight the biomass plants in the energy mix, this scenario has been divided into two. The normal median scenario takes into account all power plants including projects without considering the environmental

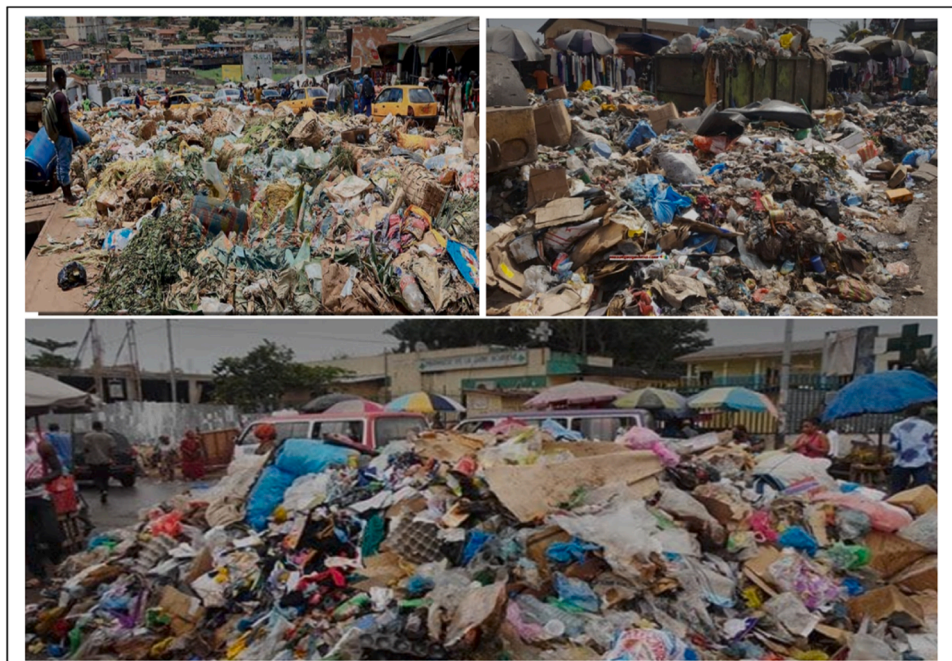


Fig. 1. Garbage collection areas of 3 markets in Yaounde.

impacts. The median sustainable scenario recommends the progressive replacement of some fuel power plants with biomass power plants to reduce the emissions of GHG. A comparison of the two scenarios is made to determine the beneficial expansion plan for meeting future energy needs, and also to determine the amount of GHGs that can be avoided in each system by meeting the demand. This study brings fundamental information for the electricity sector in Cameroon and the sustainable development of the country: (1) a model for electricity generation is proposed taking into account the valorization of biomass; (2) the environmental impacts on the production of electricity if the limitation of GHG is taken into account for two developments scenario is presented; (3) the quantities of GHGs that the country will avoid emitting during the production of electricity in Cameroon for the RIN and RIE are estimated. In this study, the issue of electricity generation in Cameroon is considered separately for the three interconnected systems as the situation stands. Existing studies have considered the supply system to be global, but this approach does not take the current reality of the three networks into account. The frequent blackouts of electricity in the country are also caused by the problem of electricity transmission. Biomass power plants presented in this paper would solve this problem through local production. The first part of this article is devoted to the methodology, an analysis of past and future energy demand is then performed based on the PDSE. The second part presents the data collection methodology and the IAEA modeling tool used in this article. The last part provides an analysis and discussion of the results obtained.

2. Problem statement

Global energy demand is increasing over time. Currently, much of the increase in demand is coming from developing countries. Cameroon has been experiencing a deep energy crisis for several years. This is due to demographic and industrial growth. In the specific case of electricity, load shedding and the lack of quality energy (low voltage) for industries are the consequences of this crisis. This deficit affects areas such as health, education, industry, and social services. Rural areas and some parts of the city do not have access to electricity. But this situation is a paradox because the country has considerable energy potential.

In Cameroon, the size of garbage collection areas sometimes exceeds 200 m² Fig. 1 and in a neighborhood of 1000–1500 households, there are between 5 and 10 collection points.

The frequency of collection of this waste remains very low. This daily waste is often burned in the open air when the only company in charge of the collection is overloaded. This situation is the source of the insalubrity which exists in some cities of the country. But this available waste is still subject to bad practices that generate GHGs, for example burning in open air. The conversion of this waste to generate electricity will contribute to solving the problem of insalubrity and energy demand.

Electricity in Cameroon comes mainly from hydroelectric plants (57%). These have very high installation costs. Thermal power plants using fuel oil emit a lot of greenhouse gases. They endanger the population because in Cameroon they are often installed not far from the houses. Also, the inequality in the distribution of hydroelectric power plants, and the problems related to the transport of electricity (very high costs of construction of lines) justify the need to seek new and decentralized energy solutions.

3. Literature review

From 1998–2017, several topics and works in the research and use of biomass energy have been addressed. These topics also refer to environmental protection (Guozhu et al., 2018). The work contains 9514 literature reports in the Core Collection of the Web of Science with the keywords "Biomass energy".

Several similar works of some authors have caught our attention and have been important for this study. For example, Qiang et al. (2009) proposed an energy planning model for China. The analysis of this model

(MESSAGE-China) was used to evaluate the entry trend of new power technologies and their contribution to GHG mitigation in the Chinese energy mix. The result expected from this work showed that traditional renewable technologies, high-efficiency coal-fired power generation, and nuclear power contribute significantly to GHG mitigation in the near term. In addition, solar, biomass, and carbon capture and storage are medium- and long-term options. They note that some plants have low capital costs and low emission factors. To better understand the energy transition between coal and biomass in China, Changbo et al. (2019) proposed a framework for full economic cost accounting and estimated the direct and external costs of coal and biomass power generation in China.

The work done by Subhash et al. (2010) and Daniel et al. (1993) showed that the activity of electricity generation produces greenhouse gases. According to the studies of Subhash et al. (2010), 40% of carbon emissions come from power sector activities. These gases are largely responsible for observed global climate change and harm biodiversity (IPCC, 2007). According to Solomon et al. (2009), the level of development and economy of a country is affected if it has a very high GHG emission power. At the moment, all countries are engaged in the fight against climate change which has been ignored for a long time. These countries are developing techniques to reduce dependence on fossil fuels. Finding concrete solutions is a challenge that all countries have to confront. Countries like the United States, China, Morocco, and even Cameroon are developing several projects to develop renewable energy. In the same direction, the European Union has made significant progress toward its 2020 renewable energy target (Nicolae Scarlat et al., 2016). Similar studies have been conducted in Cameroon by Guemene Dountio et al. (2016) to determine the overall emissions from the energy sector in Cameroon.

Biomass is considered to be one of the most easily available and high-potential renewable power sources (Glushkov et al., 2021). According to this study, it has been established that the combustion of biomass mixtures has better environmental characteristics. Pyrolysis of a biomass mixture from a group of components has a 30% higher yield of combustible gases (Glushkov et al., 2021). This study also showed that the biomass obtained after a conversion process has a higher yield than that of a traditional process.

A country like Brazil has seized the opportunity of renewable energy production. Developing countries will have an interest in developing themes of high relevance in the context of climate change. (Ana Pimenta et al., 2016) explore electricity generation in Brazil and investigate the potential for sustainable energy production from biomass. This study concludes that biomass in Brazil has the potential to grow as a renewable and clean energy source. This potential can be explored by conserving the environment and encouraging the creation of decentralized systems, making Brazil a key player in climate change objectives in the coming years. In Malawi, a similar study was carried out by Franziska et al. (2018). The study suggests policies for sustainable biomass development to improve energy supply and food security at the same time.

Self-generated electricity using mixed biomass compost is possible Muhammad (Naqvi et al., 2017). This is produced by gasification. This work presents four scenarios and concludes that the cost of producing electricity from biomass varies between 29US cents/kWh and 40US cents/kWh depending on the plant load and capacity.

According to a study carried out in 2016, Biomass has historically been the world's main source of energy, primarily for cooking, heating, and small cottage industries. It is experiencing a strong revival and is currently used in a multitude of applications. It provides between 10% and 14% of the world's energy, with a potential of 30–40%, and is an integral part of all energy scenarios (Rosillo-Calle, 2016). Fossil fuels are being replaced with renewable energy resources (biomass and biomass waste, solar, geothermal, wind, etc.) to ensure sustainable development, reduce the dependence on fossil fuels, and address environmental challenges including climate change. Today, biomass produces 5×10^{19} kJ of energy/per year, which corresponds to 10% of the annual global

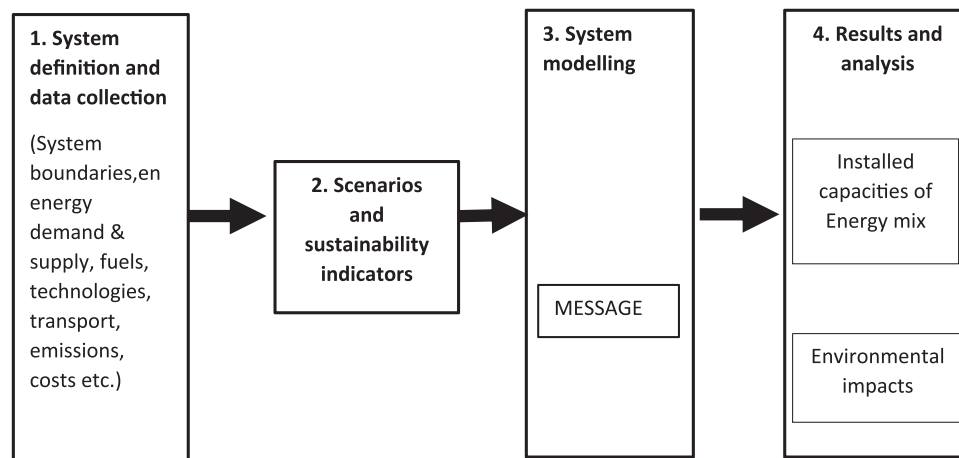


Fig. 2. Overview of the methodology used to evaluate the environmental impacts of the electricity system in Cameroon.

Table 6

Projected energy source prices (AIE, 2007).

Fuels	Prices 2010 US\$2005 per tonne	Prices 2020 US\$2005 per tonne	Prices 2030 US\$2005 per tonne	Annual price growth rate (%), EIA (2006)
Natural gas	191	211	233	1.1
Diesel (international)	475	567	678	1.8
Diesel (domestique)	355	424	507	1.8
Biomass (fuel wood)*	25	25	25	0.1
Coal* *	42	43	43	0.1

* * In Cameroon two types of biomass can be used. Woody biomass and household and urban waste. Their costs depend on the distance between the collection areas and the potential treatment or recovery sites.

* In Cameroon, no power plant has coal as its source.

Table 7

Projected technology prices (EIA, 2007).

Technology type	Investment costs (US\$2005 per kW)	Fixed costs (US\$2005 per kW)	Variable costs (US\$2005 per kWh)
Gas turbine 2020	574	11	0.003
Gas turbine 2030	561	11	0.003
Gas CCGT-2020	845	11.8	0.003
Gas CCGT-2030	734	11.8	0.005
Steam turbine-2020	405	11	0.005
Steam turbine-2030	399	11	0.005
Coal 2010	830	8	0.004
Coal 2030	800	8	0.004
Biomass 2020	1721	50	0.008
Biomass 2030	1534	50	0.008
Solar thermal 2020	2309	53	-
Solar thermal 2030	2067	53	-
Solar PV 2020	3923	11	-
Solar PV 2030	3569	11	-
Large hydro 2020	1500	13	0.0005
Large hydro 2030	1426	13	0.0005
Small hydro 2020	1980	12	-

energy consumption. Considering the variety of biomass resources, this value is predicted to reach 150×10^{19} kJ by 2050. Biomass may become even more important for use as an energy resource and chemical raw material in the 21st century. Hydrothermal biomass conversion stands out as a promising alternative technology (Alper et al., 2020). Alper and al. describe all the steps of biomass processing, explaining the chemistry taking place in the background of the processes in a comprehensive manner in their review. They also demonstrated that biomass could

become even more important to be used as an energy resource and chemical raw material in the 21st century. Also, the hydrothermal conversion of biomass is a promising and alternative technology (Alper et al., 2020). With the aim of improving the efficiency of biomass conversion processes, Aihua et al. (2021) studied the influence of component interaction on the pyrolysis and explosion of biomass dust. Their results show that a higher fraction of cellulose enhances the mixture's weight loss peak intensity and explosion pressure. For gasification, Zhenjuan et al. (2021) present a global economic analysis and multi-objective optimization of an integrated gasification electricity production cycle. The authors define economic and extra-economic criteria to evaluate the feasibility of the system in terms of investment and construction. The system studied was analyzed by developing a precise model in the Engineering Equation Solver software. In this study, the results show that the system has an energy efficiency of 42.7%, with an electricity production capacity of 7.768 MW. This confirms the feasibility of gasification power plants.

A more recent study was also done by Muhammed et al. (2020), with the aim of encouraging the use of renewable energies in Pakistan. Therefore, this study aimed to examine the factors influencing consumer intention to utilize renewable energy (RE) (Muhammad et al., 2021). On the other hand, authors like Lund believe that modeling renewable energy systems is 100% beneficial (Henrik Lund, 2014). He defines renewable energies, gives the advantages of their use for sustainable development, and finally the political reasons for using RE (Henrik Lund, 2014).

4. The energy situation in Cameroon

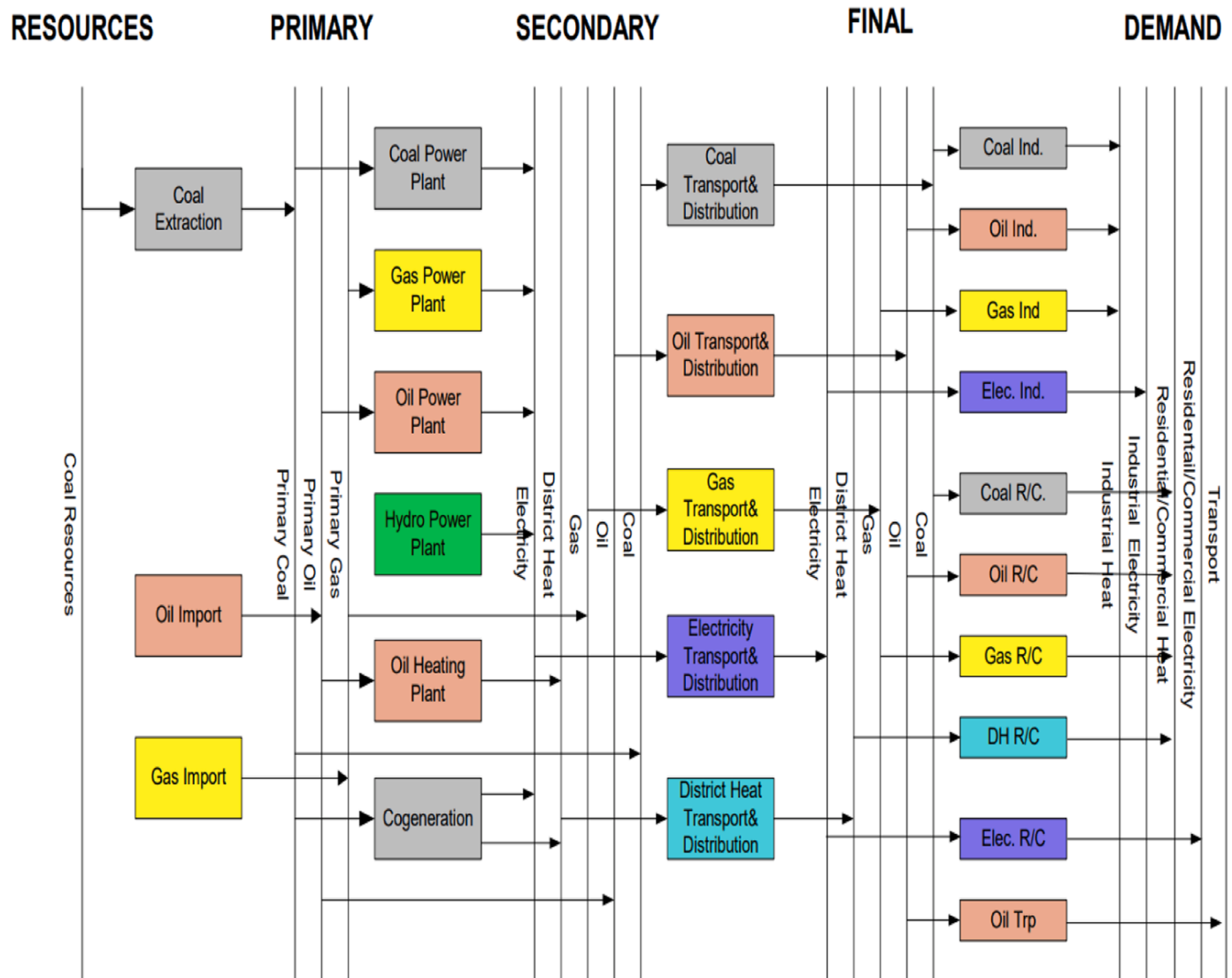
Cameroon is located in Central Africa at the bottom of the Gulf of Guinea. The surface area of the country is 475 650 km². It is located between 1° and 13° North latitude and between 8° and 17° East longitude (Guemene Dountio et al., 2016). Its strategic position gives it the

Table 8

Evolution of the structure of Cameroon's GDP in the Median economic scenario.

GDP structure		2012	2015	2017	2020	2025	2030	2035
GDP growth rate	%	-	5,50	5,60	5,70	6,00	6,10	5,80
Primary share	%	36,30	36,50	38,15	34,99	33,75	32,07	30,46
Secondary share	%	22,70	22,50	20,84	22,41	23,25	23,93	24,53
Tertiary share	%	41,00	41,00	41,00	42,60	43,00	44,00	45,00
Share of industries manufacturing	%	15,30	15,40	15,60	16,10	16,30	16,80	17,20

Source: scénario de référence DSCE 2009, World Bank

**Fig. 3.** Schematic Presentation of Some Energy Chains.

advantage of having several forms of energy. Among these forms, forest biomass and hydroelectricity are the most important. But the valorization of this potential for electricity production is still very low (Ackom et al., 2013). The biomass potential comes from forestry, household waste, and food industries.

If the biomass potential is exploited to the maximum extent, electricity production from this source can contribute to 40% of the electricity demand in Cameroon (Fotsing et al., 2015). Another source of energy currently exploited for electricity production is solar photovoltaic. But this technology is more used for self-production to supply isolated households only. This production is not interconnected to any

network. Cameroon's energy production capacity is estimated at 2327.5 TWh (MINEE, 2006). Its hydroelectric potential is projected at 20 GW but only about 3% of this potential is exploited (Gujba et al., 2011). The problem with hydroelectricity in Cameroon is that some areas of the country do not have rivers. The supply may be great in some areas, but there is a problem with the transmission of electricity. Despite this energy potential, the electricity demand still exceeds the supply. The energy situation of the country is included in Table 1 and the CO₂ emission from each energy source is in Table 2.

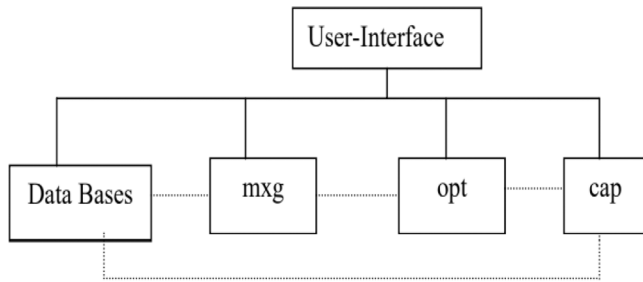


Fig. 4. Interrelationship between MESSAGE components.

5. Methodology

The methodology for obtaining the energy mix planning and the environmental impact assessment for the case of Cameroon is described in the diagram in Fig. 2. The same methodology has been used (Gujba et al., 2011) with the only difference being that the modeling tool used in this study is MESSAGE (Model for Energy Supply Strategy Alternatives and their General Environmental Impact).

5.1. Data collection

For this study, the data used for the realization of this model come mainly from surveys, from the National Institute of Statistics, the body responsible for providing certain data and statistics to researchers; various headquarters of existing power plants, the World Bank, various framework documents for the electricity sector in Cameroon. For the assessment of energy demand, data come from the Ministry of Water and Energy (MINEE, 2006). The study was carried out and recorded in document PDSE2035. It provides the demand for the three

interconnected networks. Data relating to the economy, population, and any other information serving as input for the modeling tool were provided by the National Institute of Statistics (INS) which is the body responsible for carrying out certain inquiries for the drafting of certain reports. All data relating to the power plants provided by MINEE, the structure in charge of the distribution and transport of electricity, ENEO, and the various reports of the development plans for the electricity sector.

The costs of the technologies and the various fuels are taken from the projection document published by the EIA (Energy Information Administration) whose price references by the form of energy and by type of turbine used are shown in Table 6 and Table 7 respectively.

Based on these characteristics, the technological base of existing and planned power plants in Cameroon has been established and given in Table A.1 and Table A.2 in the appendix. In these tables are also found the meanings of the abbreviations of the technologies which will be found in the section reserved for the results.

5.2. Presentation of scenario

The formulation of scenarios is the first phase to design for this work. A scenario in our context is a set of socio-economic and technological parameters linked together and presenting a global vision of a system (in our case Cameroon).

Therefore, several international and national organizations have predicted the future of the Cameroonian economy. These are in particular the World Bank, the FIM, and the government's Vision 2035 (SIE; 2011). However, this study is only based on the median scenario. The highlights of this scenario are:

1. High economic growth but lower than that necessary for the perspective of an emerging Cameroon in 2035. The growth rate will

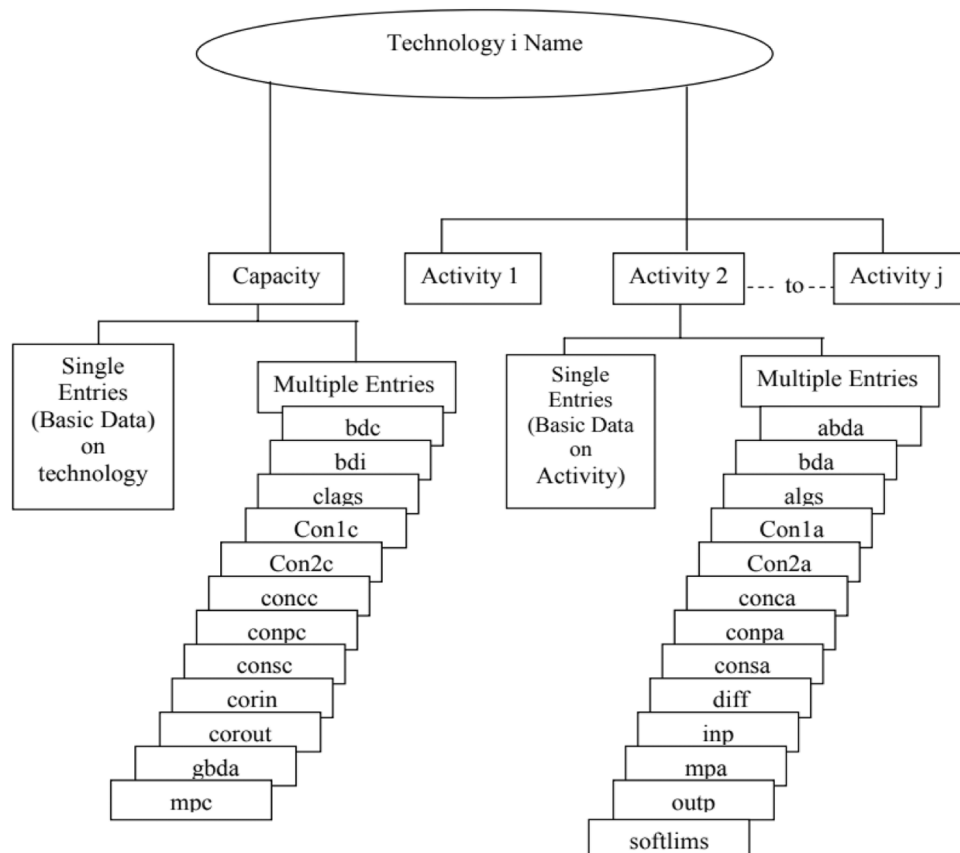


Fig. 5. Structure of data on a technology.

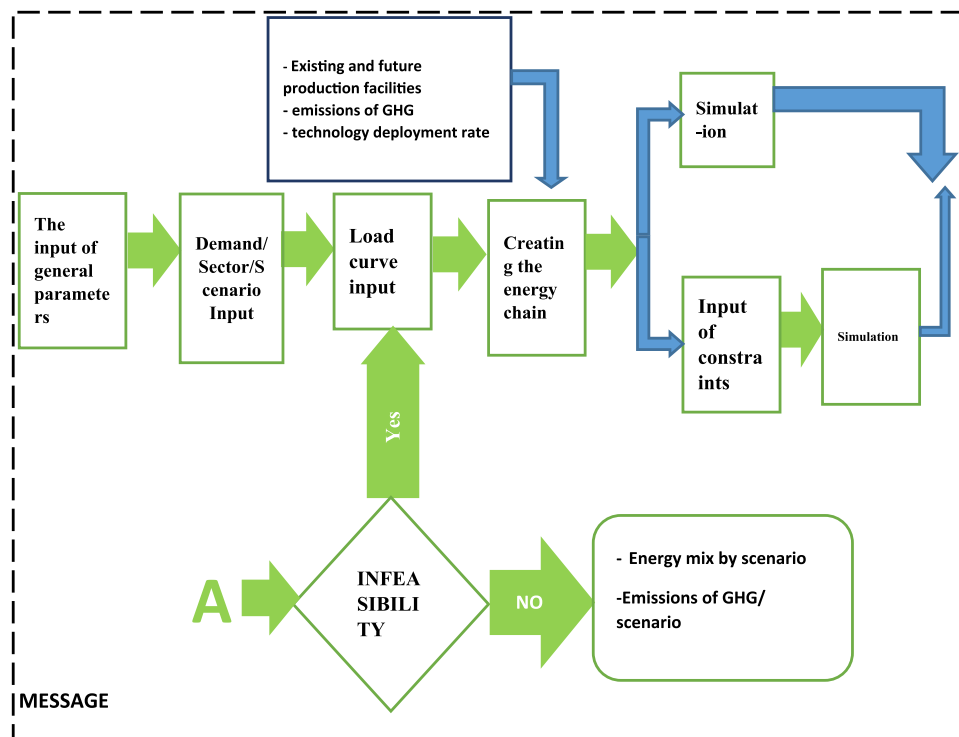


Fig. 6. logical structure of Methodology.

painfully reach the 6% mark, far from the double-digit growth rate necessary to reach emergence objectives.

2. The slow and gradual change in the structure of the economy. Manufacturing industries are developing slowly without much support from public investment, and the modernization of agriculture is slow but steady. The service sector is struggling to professionalize and is still marked by a strong dominance of commercial activities.
3. The improvement in the feeling of growth is slight, the average household income has increased slightly over the years. The data resulting from this scenario have been recorded in [Table 8](#).

The figures and tables in the appendix present other types of data used and the energy chain is presented in [Figure A.1](#) in this study. The energy map presented in [Figure A.2](#) locates some power plants and dams in the country.

Electricity generated from biomass is some time produced by direct combustion. Biomass is burned in a boiler to produce high-pressure steam. In thermal power plants, steam is produced from fossil fuels or nuclear energy. In biomass power plants, the heat used to produce steam comes from combustion, gasification or pyrolysis. To generate biomass power plant in MESSAGE, the input are: the quantity of biomass, the efficiency of the power plant, the capacity of the plant, the investment cost, the variable costs, the time live.

5.3. Analysis framework MESSAGE

MESSAGE is an acronym for Model for Energy Supply Strategy Alternatives and their General Environment impacts. MESSAGE was designed by Professors Wolf Haeefe and Alan S. Manne for the "global energy project" conducted at the International Institute for Applied Systems Analysis (IIASA) in the 1970 s (Messner et al., 1995) and (IEA, 2014). It was designed to formulate and evaluate alternative energy supply strategies using constraints on investment, market penetration of new technologies, availability of petroleum resources, and environmental protection. The principle of its modeling is the optimization of an objective function (lower cost, lower environmental impact, maximum

self-sufficiency.) using the definition of constraints. MESSAGE is based on the technical and economic description of an energy system. This includes the definition of forms of energy (primary energy, demand, secondary energy, etc.), fuels, and associated technologies (electricity, diesel, ethanol, coal, etc.).

This software has already been used by Guemene et al., to assess the overall energy supply of Cameroon (Guemene Dountio et al., 2016). It was also used here to assess the energy supply of each interconnected network. Then the greenhouse gas balance for each of these networks was done.

The underlying principle of a model, built using the MESSAGE, is optimization of an objective function under a set of constraints that define the feasible region containing all possible solutions of the problem. The value of the objective function helps to choose the solution considered best according to the criteria specified. In general categorization, models built using the MESSAGE belong to the class of mixed integer programming models as they may contain some integer variables. A set of standard solvers (e.g. `glpk` and `cplex`) can be used to solve these models. The main objective of developing the MESSAGE software, however, was to facilitate the building of an energy system model.

An energy model is designed to formulate and evaluate alternative energy supply strategies consonant with user-defined constraints such as limits on new investment, fuel availability and trade, environmental regulations, and market penetration rates for new technologies. Environmental aspects can be analyzed by accounting, and if necessary limiting, the amounts of pollutants emitted by various technologies at various steps in energy supplies. This helps to evaluate the impact of environmental regulations on energy system development.

In the first approximation, a model, built using the MESSAGE, could be labeled as a physical flow model. Given a vector of demands for specified energy goods or services, it assures sufficient supply, utilizing the technologies and resources considered. MESSAGE allows modeling of all steps in the energy flows from supply to demand, which is generally referred to as the energy chain, and steps are called levels. Fig. 3 shows the schematic representation of some energy chains.

To sum up, to build a model with MESSAGE, you need to choose a

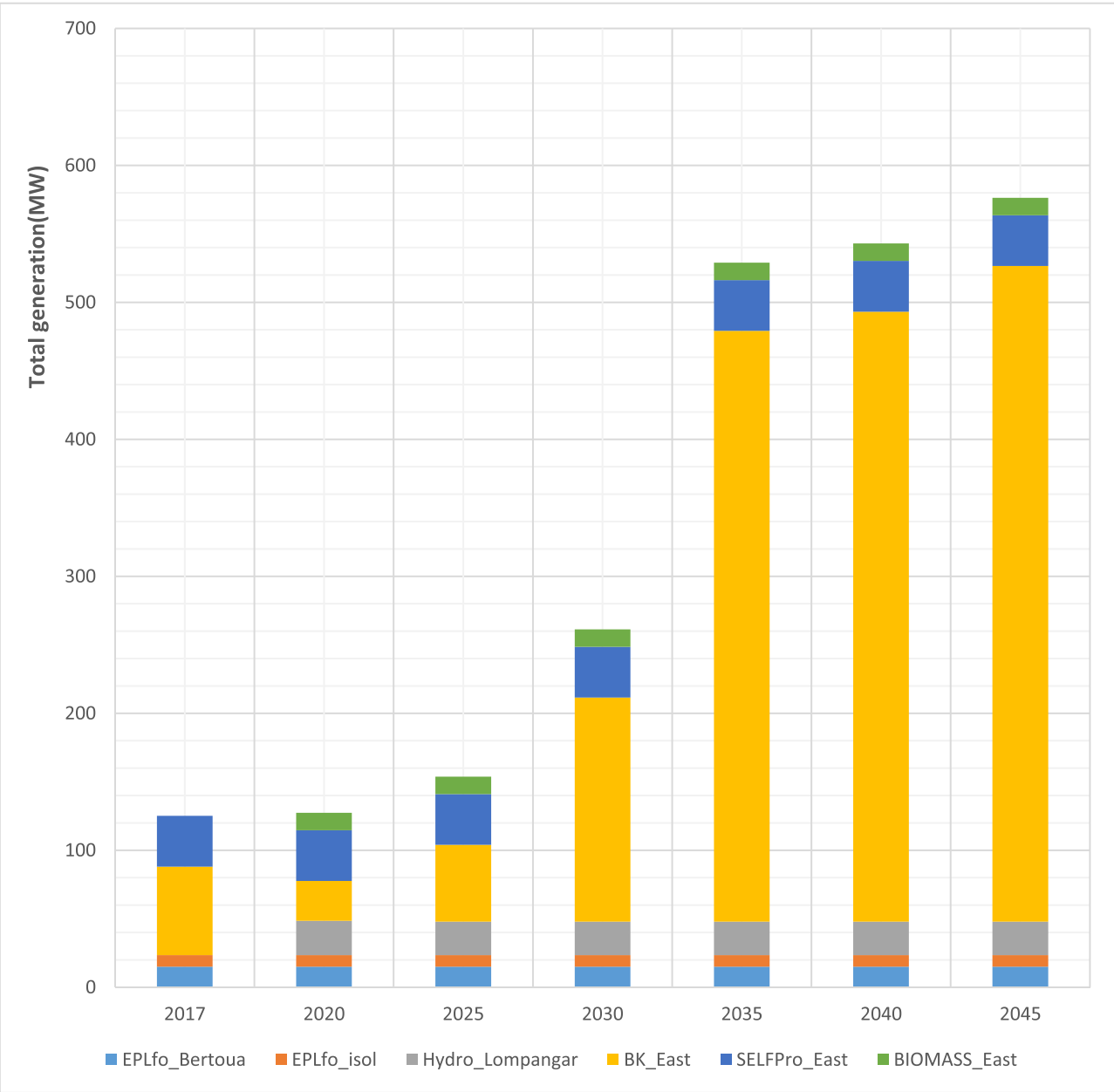


Fig. 7. Power generation and mix in the baseline scenario RIE.

base year. The base year is the year in which the energy mix data is available. Define the scenarios and enter the equivalent energy demand. The model is then created based on a technical and economic description of the energy system. These include the definition of energy forms, and associated technologies. The model evaluated the quantity of GHG emitted using existing power plants. This data include the activity and capacity of the power plants. Power plants are defined by their inputs and outputs, their efficiency, and their variability if they have more than one input or output. These parameters include investment costs, operating costs, input energy import costs, etc. Year of construction, life time, output, installed capacity. In this study, the country's electricity demand is not supplied. In the model, the backup power plant was introduced. This plant is the main focus of analysis and action for the future energy mix.

5.3.1. MESSAGE software

The current version of the MESSAGE software consists of the following components.

- A user interface for building a model.
- Databases.
- A matrix generation program called “mxg”.
- An Optimization program called “opts”.
- A program for the post processing of the solution for extraction of results called “cap”.

Fig. 4 shows the flow of control and information between these components in the execution of the MESSAGE program.

5.3.1.1. Database management. The MESSAGE program maintains, mainly, four types of databases for storing all input data required for a model developed in a case study. These four types of databases are as follows.

- (1) Technology database, called tdb, is created and maintained independently of any case study.

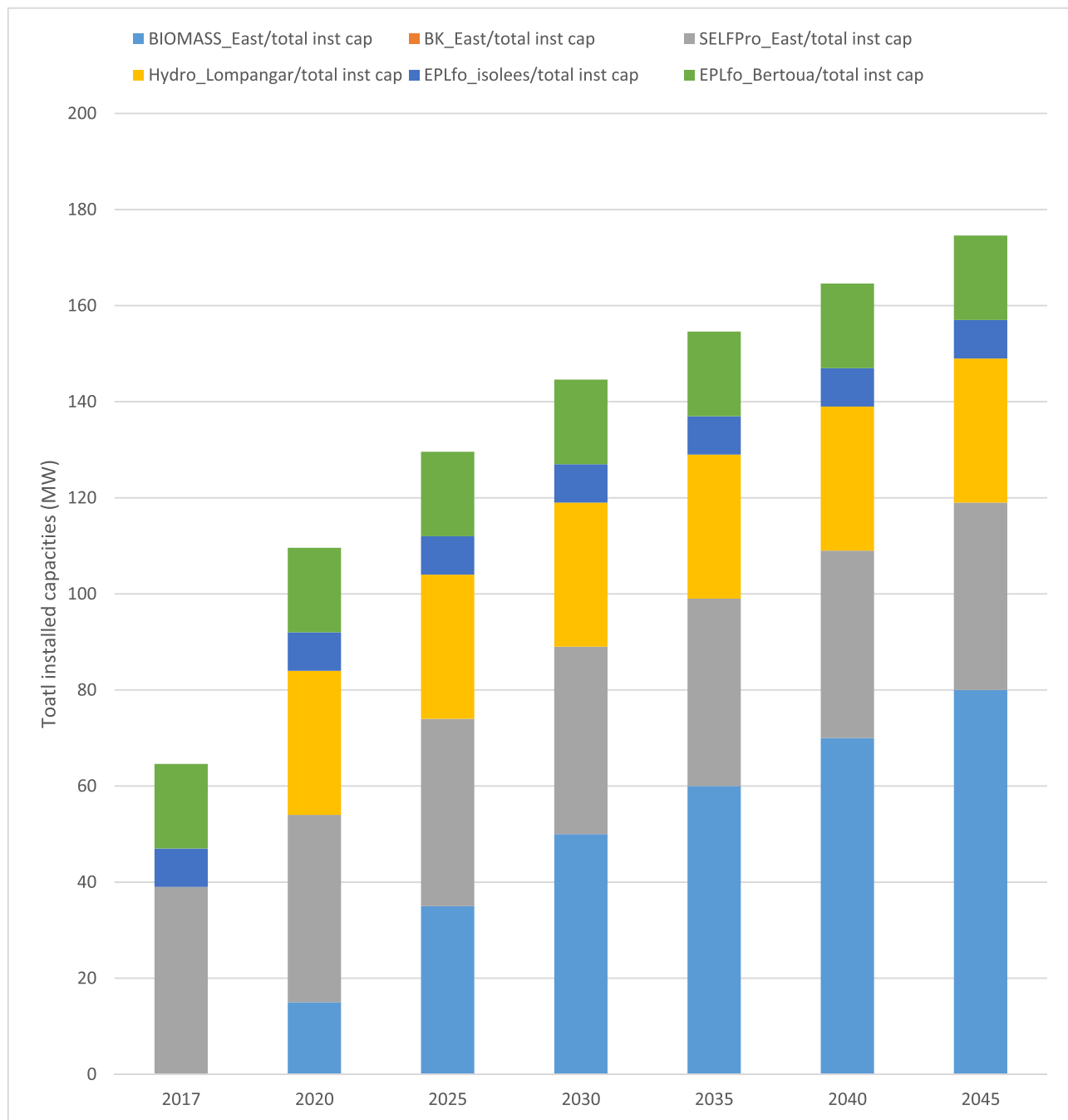


Fig. 8. Evolution of installed capacity of the RIE in baseline scenario.

- (2) Application database, called adb, is created and maintained for each case study.
- (3) Local database, called ldb, is created and maintained for each scenario in a case study.
- (4) Update database, called upd, is created and maintained to update values in the adb and/or a set of scenarios in a case study across the board. Fig. 5 shows the structure of this component, which is exactly the same in all four databases.

5.3.2. Some mathematical formulations of MESSAGE

a. The contribution of capacities existing in the base year

The possible contribution of an installation that exists in the base

year is kept track of over time. There are two possibilities to give the necessary information to MESSAGE.

1. Define the capacities that were built in the years iy_r , $iy_r - \tau + 1$, with iy_r = base year and τ = plant life in years explicitly. These capacities are then distributed to historic periods of the length ν .
2. Define the total capacity, c_o , that exists in iy_r and the rate at that it grew in the last τ years, γ . This information is then converted to one similar to Eq.1 by using the function:

$$y_o = c_o \frac{\gamma^{-\nu} - 1}{\nu(\gamma^{-\tau} - 1)} \quad (1)$$

$$y_t = y_o * \gamma^{-t/\nu}, t = 1(1)\frac{\tau}{\nu} \quad (2)$$

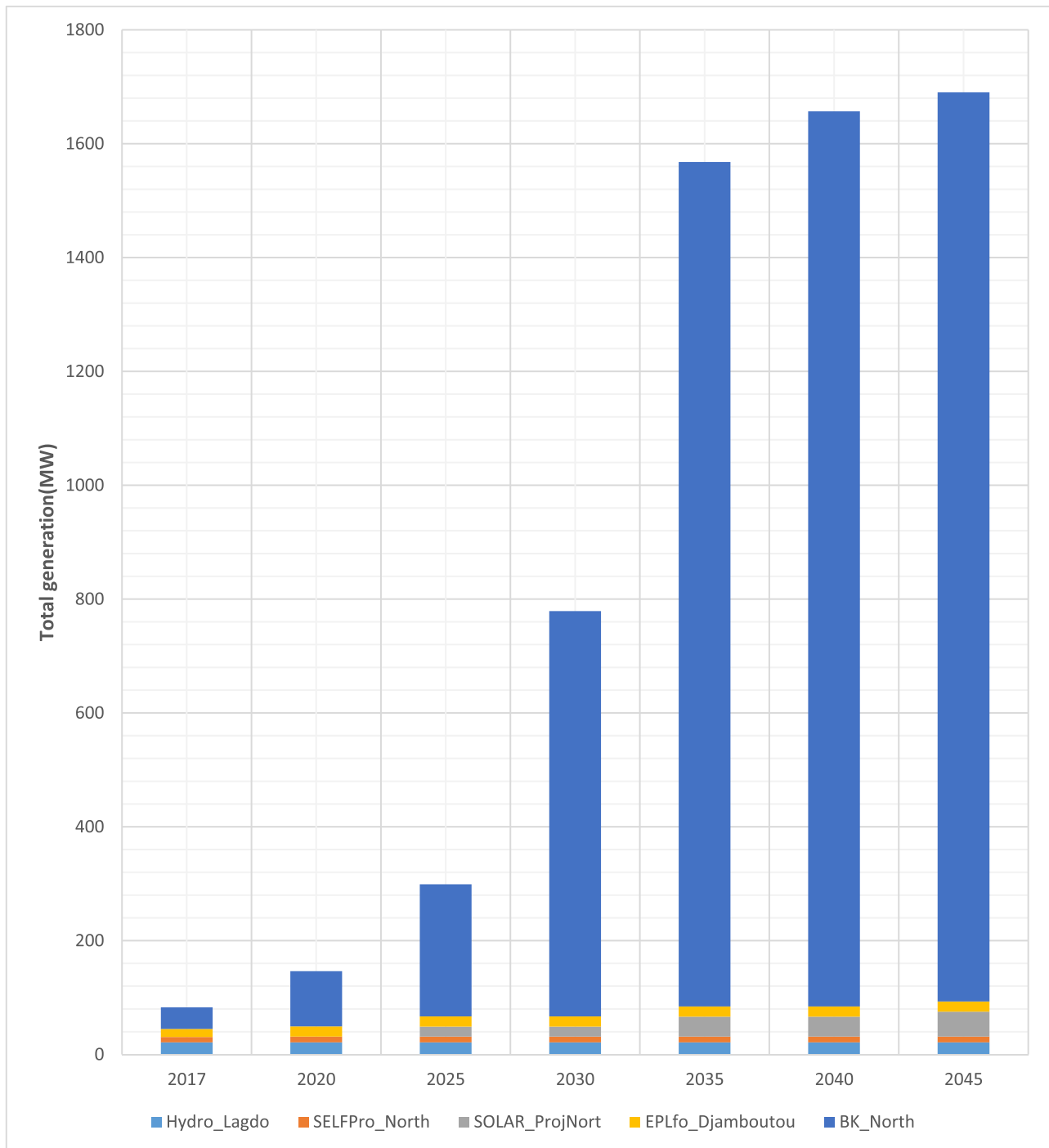


Fig. 9. Evolution of electrical production in the RIN for the baseline median scenario.

Where,

y_t Is the annual construction in the period t , (0 = base year),

γ Is the annual growth of new installations before the base year,

c_o Is the total capacity in the base year,

τ Is the plant life, and,

ν Is the length of the periods in that the time before the base year is divided.

The right-hand sides in the capacity constraints are derived by summing up all the old capacities that still exist in a certain period (according to the plant life). If the life of a technology expires within a period, MESSAGE takes the average production capacity in this period as installed capacity (this represents a linear interpolation between the

starting points of this and the following period).

a) Capacities that operate longer than the time horizon

If a capacity of a technology is built in one of the last periods its lifetime can exceed the calculation horizon. This fact is taken care of reducing the investment costs by the following formula:

$$C_t^r = C_t * \frac{\sum_{k=1}^{\tau_p - \nu} \prod_{\tau=t}^{t+k-1} \frac{1}{1+dr_\tau}}{\sum_{k=1}^{\tau} \prod_{\tau=t}^{t+k-1} \frac{1}{1+dr_\tau}} \quad (3)$$

Where,

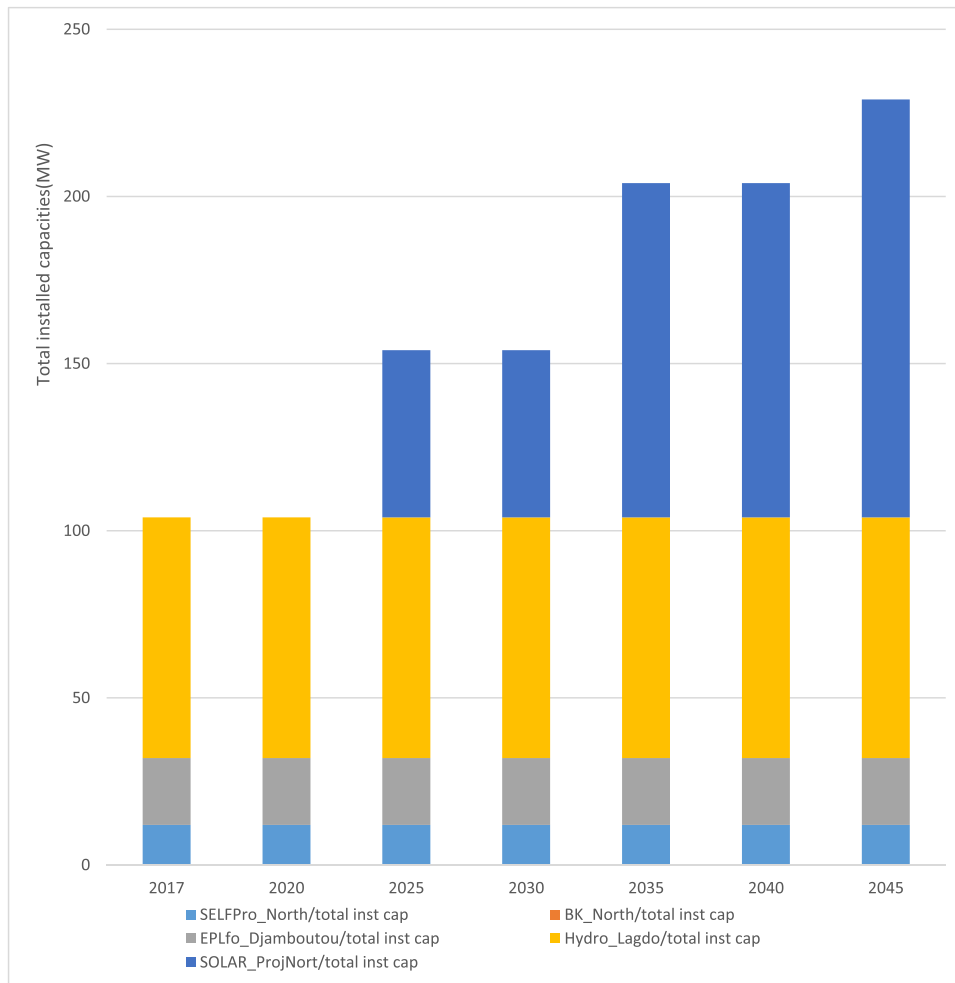


Fig. 10. Evolution of the total capacities to be installed in the RIN for the baseline median scenario.

ν Is the number of years the technology exists after the end of the calculation horizon,

dr_t Is the discount rate for year t ,

τ_p Is the plant life in years,

C_t Is the investment cost in year t , and

C_t^r Is the reduced investment.

b) Own-price elasticities of demand

The costs and levels of demand reduction can be derived from the investments and savings that are associated with certain additional installations, like, e.g., three-glass windows to save space heating. The form of the own price elasticity function of demand is:

$$\frac{Q}{Q_r} = \left[\frac{P}{P_r} \right]^\epsilon \quad (4)$$

Where,

Q_r Is the reference demand level,

P_r Is the reference price level, and

ϵ Is the elasticity, (assumed to be < 0).

It says that the demand will decrease by a factor of x^ϵ if the price rises by x . This function is approximated by a step-function of the following form:

The demands (Q) and prices (P) are normalized to the reference levels given by Eq. 5 and Eq. 6:

$$Q = q * Q_r \quad (5)$$

and

$$P = p * P_r \quad (6)$$

The normalized values follow the function

$$q = p^\epsilon \quad (7)$$

Or

$$p(q) = q^{\frac{1}{\epsilon}} \quad (8)$$

To reduce the demand to the level q_i the supply has to have the cost

$$c(q_i) = \int_{q_i}^1 q^{\frac{1}{\epsilon}} dq = \frac{1}{1 + \frac{1}{\epsilon}} * \left[1 - q_i^{1 + \frac{1}{\epsilon}} \right], \quad (9)$$

a function increasing monotonously with decreasing q_i . In absolute terms, this means that the cost would be higher by an absolute value of

$$R(Q_i) = c\left(\frac{Q_i}{Q_r}\right) * Q_r * P_r \quad (10)$$

compared to the cost at the reference level.

The step-function is then defined by choosing certain levels of demands and prices (Q_i, P_i), $i = 1(1)n$, with $Q_i < Q_r$, that fulfill the elasticity function. The code can choose, which demand level it supplies, but if it supplies a level $Q_i < Q_r$ it has to pay additionally $R(Q_i)$, the cost of reducing the demand to level i .

• Technologies without Load Regions

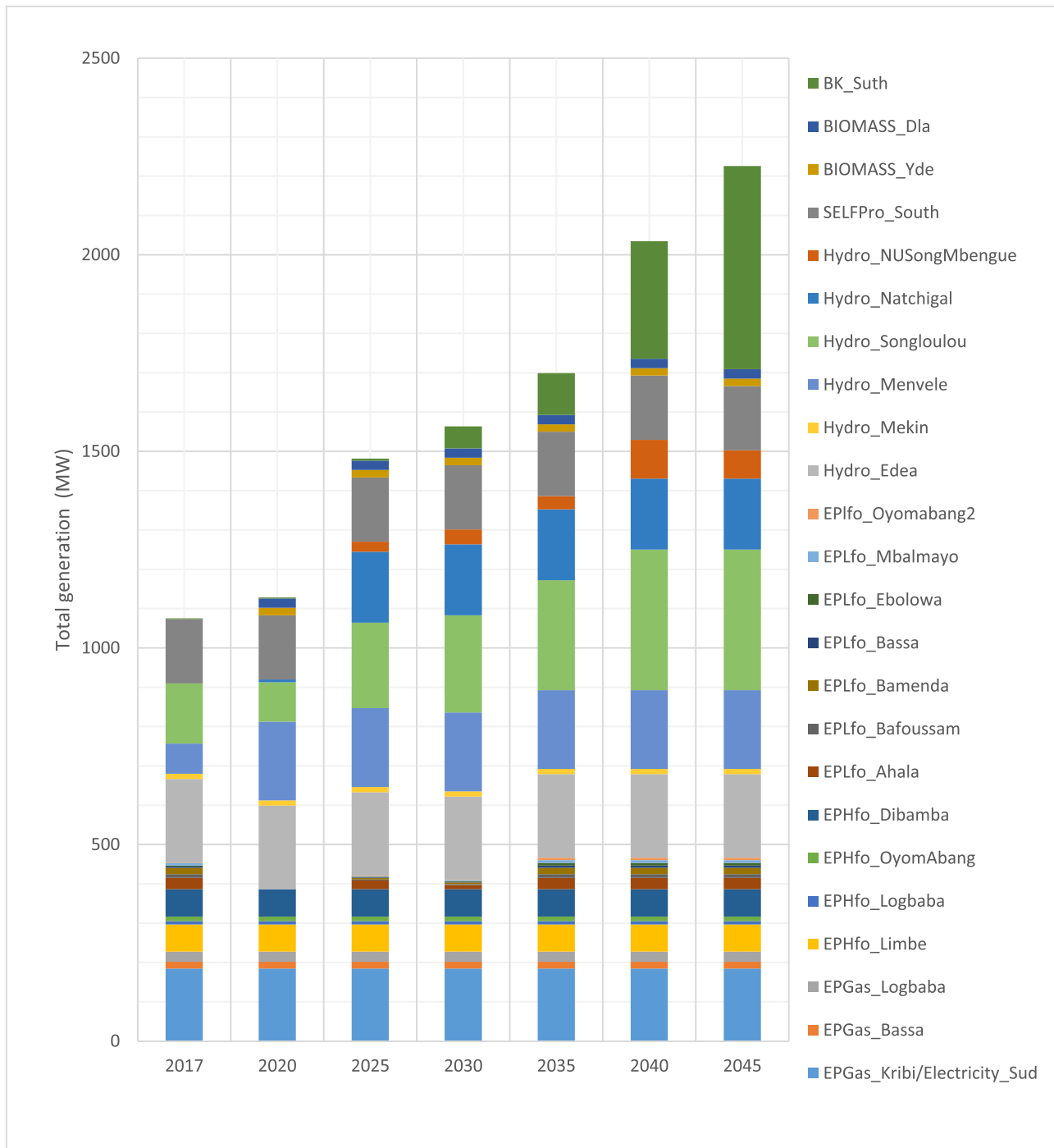


Fig. 11. Evolution of electrical production in RIS for the baseline median scenario.

$$\epsilon_{svd} \times z_{svd} \dots t - \sum_{\tau=t-\tau_{svd}}^{\min(t, k_{svd})} \Delta(\tau-1) \times \pi_{svd} \times f_i \times Y_{zsvd} \dots \tau \leq hc_{svd}^t \times \pi_{svd}. \quad (11)$$

• Technologies with Load Regions and "Free" Production Pattern

$$\frac{\epsilon_{svd}}{\lambda_l} \times z_{svd} \dots t - \sum_{\tau=t-\tau_{svd}}^{\min(t, k_{svd})} \Delta(\tau-1) \times \pi_{svd} \times f_i \times Y_{zsvd} \dots \tau \leq hc_{svd}^t \times \pi_{svd}. \quad (12)$$

• Technologies with Load Regions and "Fixed" Production Pattern

$$\frac{\epsilon_{svd} \times \pi(lm, svd)}{\lambda_{lm}} \times z_{svd} \dots t - \sum_{\tau=t-\tau_{svd}}^{\min(t, k_{svd})} \Delta(\tau-1) \times \pi_{svd} \times f_i \times Y_{zsvd} \dots \tau \leq hc_{svd}^t \times \pi_{svd}. \quad (13)$$

• Technologies with Varying Inputs and Outputs

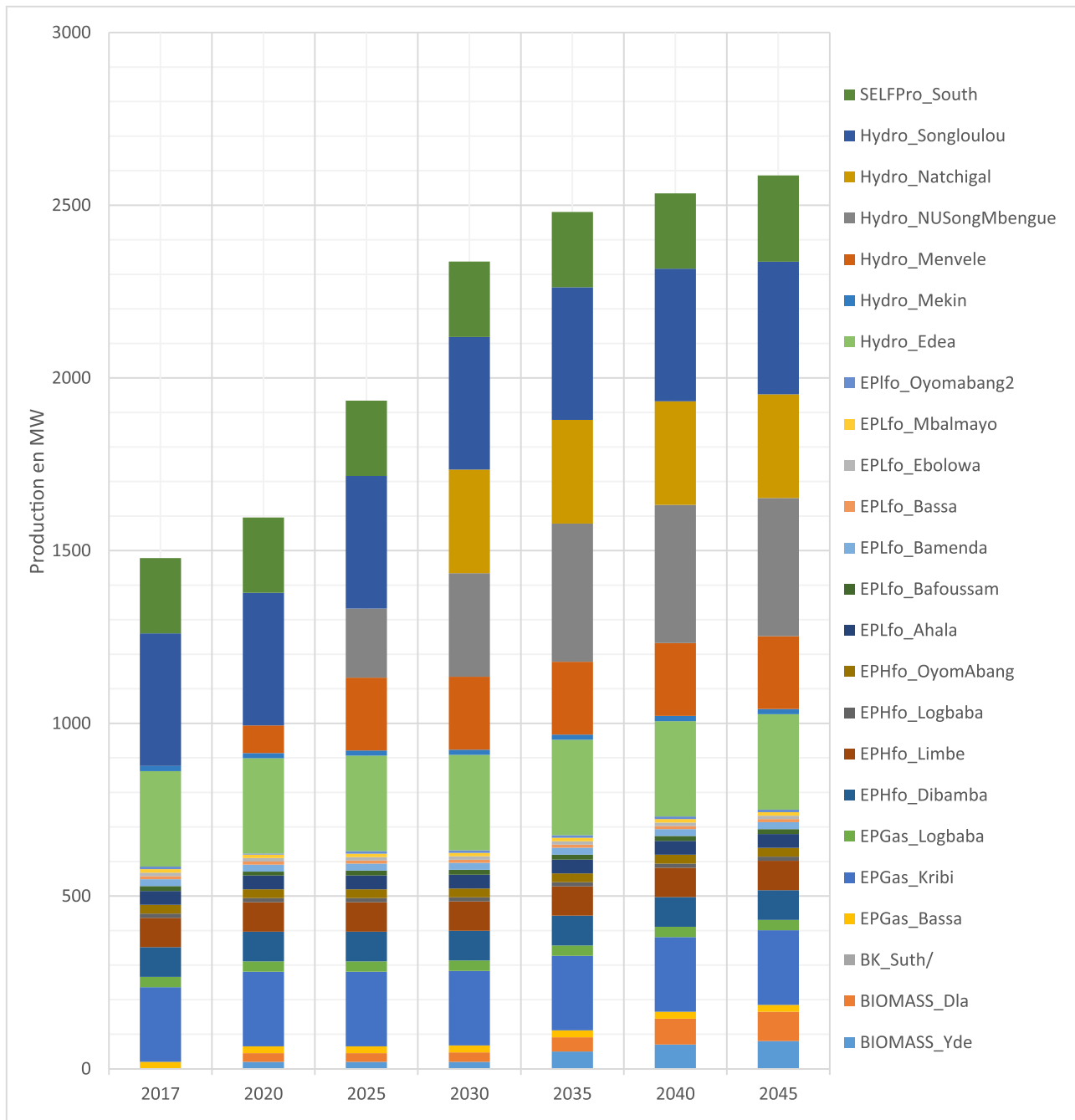


Fig. 12. Evolution of the total capacities to be installed in the RIS for the baseline median scenario.

$$\sum_{\sigma/\delta} rel_{\sigma/\delta}^{svd} \times \epsilon_{\sigma/\delta} \times z_{\sigma/\delta} \dots t - \sum_{\tau=t-\tau_{svd}}^{\min(t, k_{svd})} \Delta(\tau-1) \times \pi_{svd} \times f_i \times Y_{zsvd} \dots \tau \leq hc_{svd}^t \times \pi_{svd}. \quad (14)$$

The following notation is used in the above equations:

$z_{svd} \dots t$ Is the activity of conversion technology v in period t and, if defined so, load region l ,

$Y_{zsvd} \dots t$ Is the capacity variable of conversion technology v .

ϵ_{svd} is the efficiency of technology v in converting the main energy input, s , into the main energy output, d ,

k_{svd} Is the last period in that technology v can be constructed,

π_{svd} is the "plant factor" of technology v , having different meaning depending on the type of capacity equation applied,

$\Delta \tau$ is the length of period τ in years,

τ_{svd} Is the plant life of technology v in periods,

hc_{svd}^t Represents the installations built before the time horizon under consideration, that are still in operation in the first year of period t ,

f_i Is 1 if the capacity variable is continuous, and represents the minimum installed capacity per year (unit size) if the variable is an integer,

l_m Is the load region with maximum capacity use if the production pattern over the year is fixed,

$\pi(l_m, svd)$ Is the share of output in the load region with maximum production,

$rel_{\sigma/\delta}^{svd}$ is the relative capacity of the main output of technology (or operation mode) svd to the capacity of the main output of the alternative technology (or operation mode) σ/δ ,

λ_l Is the length of load region l as a fraction of the year, and.

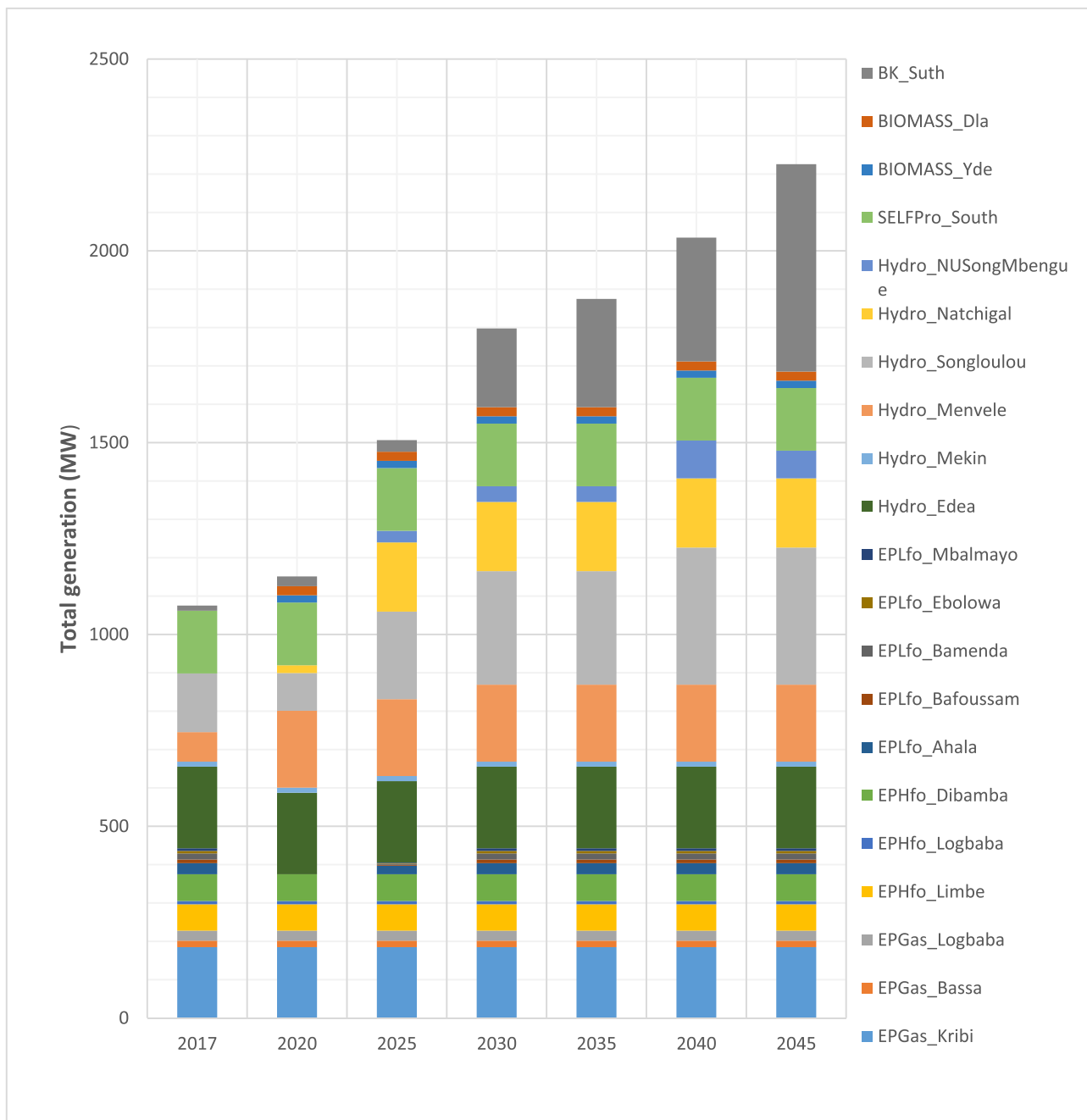


Fig. 13. RIS power generation and mix in the Sustainable scenario.

λ_{lm} Is the length of load region lm , the load region with maximum capacity requirements, as a fraction of the year.

5.4. Greenhouse gas emissions assessment

There are several methods for quantifying GHG emissions, including LCA (Life Cycle Assessment). But in this study, the GHGs were evaluated on the basic characteristics of each technology (power plant). The main GHGs that were evaluated are carbon dioxide CO_2 , methane CH_4 , and sulfur dioxide SO_2 . For each type of technology, the emission factors for each gas are entered in the constraints. There are several types of constraints, some of which set limits on emissions according to the development scenario chosen. The logical structure of this section (Methodology) is shown in Fig. 6. This work aims at consolidating the engagements taken by Cameroon at the " COP 20 " in 2014 in Lima. That

of reducing its GHG emissions by 32% between 2014 and 2035.

6. Results and discussions

The results presented below are based on two development scenarios. The median scenario without GHG limits and the scenario with GHG limits. The following subsections discuss the major results by interconnected network.

6.1. Mix of energy production in the three networks

6.1.1. Scenario without limitation of GHG, Baseline median scenario

- Results for the East Network Interconnected (RIE)

The mix of energy production in RIE in Cameroon in the baseline

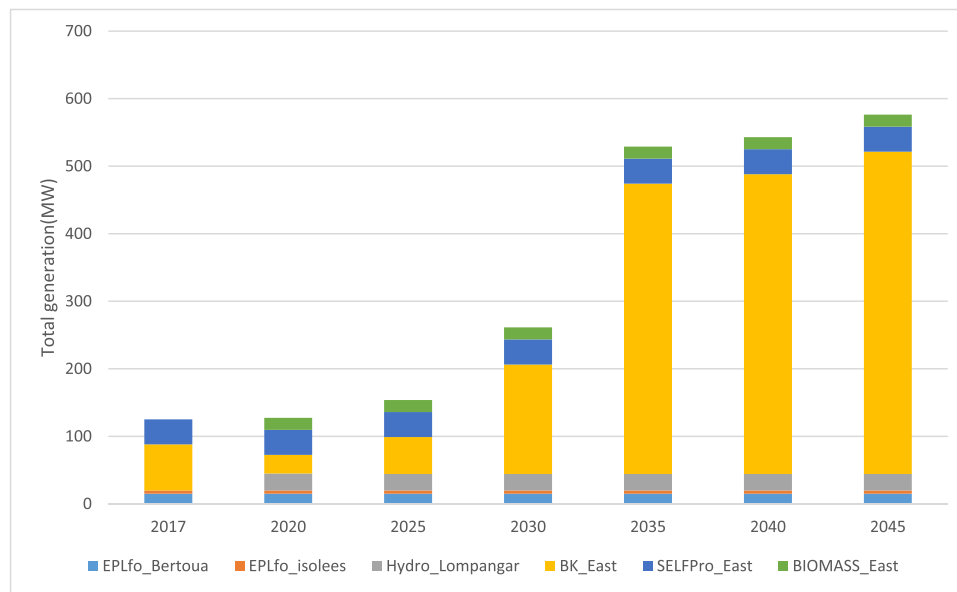


Fig. 14. RIE power generation and mix in the Sustainable scenario.

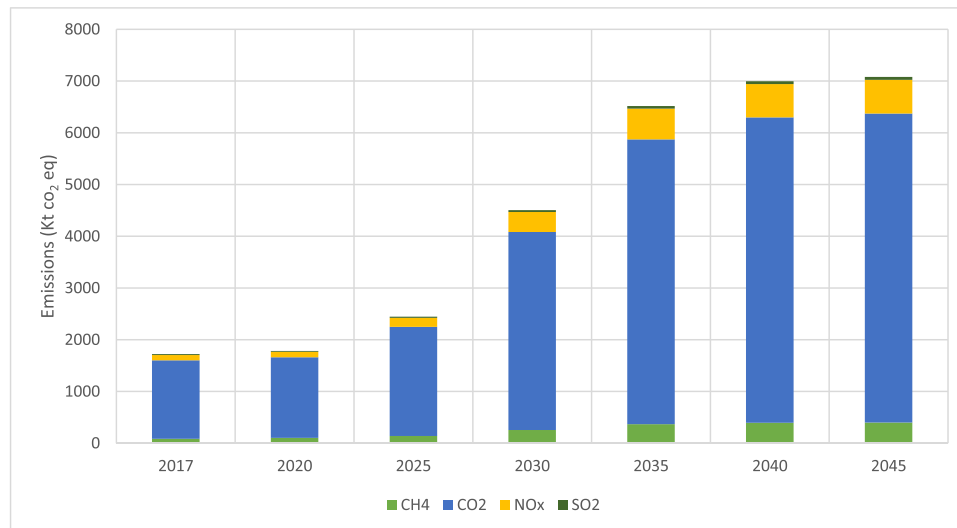


Fig. 15. Emissions of RIS, Baseline median Scenario.

median scenario is shown in Fig. 7. In that scenario, the energy production increased from 60,491 MW in 2017–97, 67 MW with a deficit that increased from 65 MW to 478 MW in the same period. It has a fairly large share of production from self-production. The network is mainly supplied by oil power stations. The commissioning into the network of the 30 MW Lom Pangar hydropower plant and the biomass power plant lowers the load shedding rate from 64.66 MW to 29 MW for the base year 2017, and this decrease can be observed throughout the period. The increase in the energy deficit represented here by the BK East technology from 2030 to 2045 is explained by the fact that, from 2030, the project to build a foundry in the Ndjoum locality of the southern border to the East region and the projects of construction industries are sources of increased demand. These industries will be connected to RIE because of the great distance between them and the RIS. The study of the availability of biomass in the East region made above, for this reason, confirms the possibility of installing biomass plants.

Fig. 8 shows the evolution of installed capacities for the RIE. This result aims at showing the policy to be undertaken in this network.

The economic and ecological solution is in favor of choosing the exploitation of the hydroelectricity potential, for example, the Lom Pangar power plant. However, this solution is not the most likely. Indeed, an increase in the capacity of the Lom Pangar power plant could have an impact on the production of other power plants installed on the same river. In order to meet the demand observed in the previous figure (Fig. 7), it would be necessary to increase the installed capacity. These installed capacities must be equivalent to those of the BK_Est technology. If this is not possible, then decision-makers must undertake other projects to build new generation units. Total electricity generation increases rapidly between 2017 and 2030 due to the addition of biomass plants to the grid energy mix.

- Results for the RIN

Fig. 9 shows the evolution of the electricity production of the RIN during the planning period. This figure shows a fairly significant increase in the energy deficit materialized here by the BK_Nord power plant. It makes the model feasible because, without it, the existing plants in the network will not be able to meet the ever-increasing demand from the year 2030. Solar_North technology

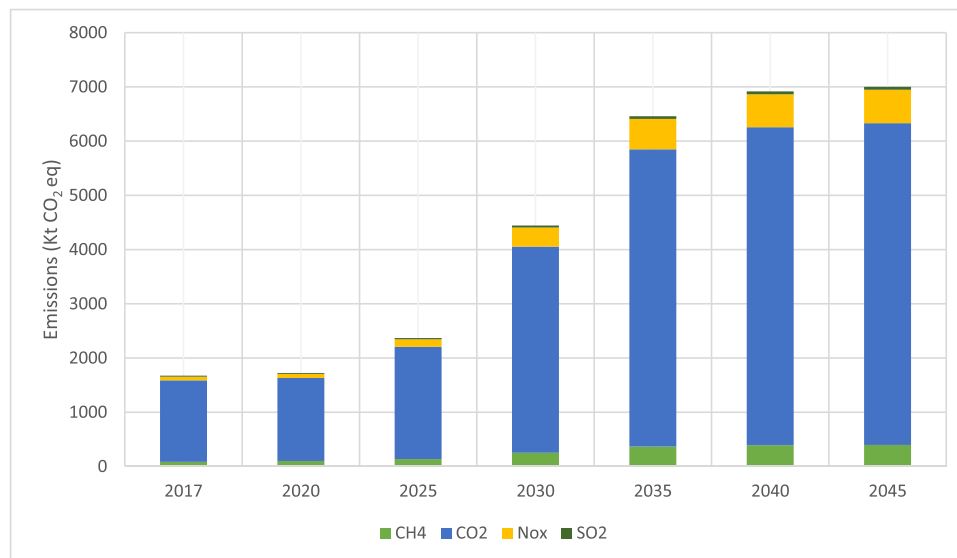


Fig. 16. Emissions of RIS, Sustainable Median Scenario.

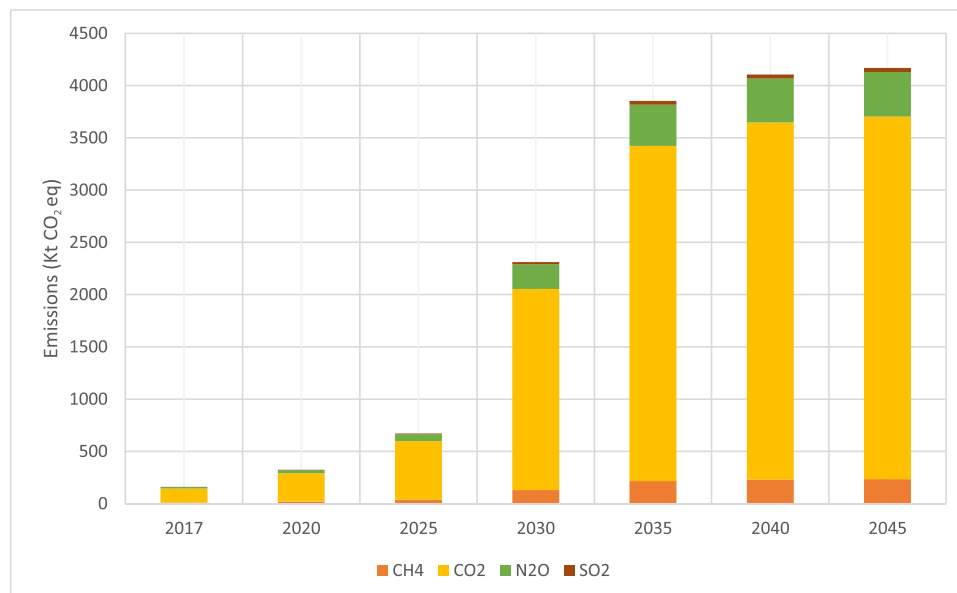


Fig. 17. Emissions of RIN, Baseline Médian Scenario.

represents the solar center which is an ENEO Cameroon project. This project promotes the construction of mini solar power plants with a total installed capacity of 35 MW by 2035. The choice of this source of energy is justified by the fact that it is the part which has the highest rate of sunshine in the country. Its participation in the RIN electricity mix will take place in 2025 according to ENEO previsions. The Lagdo hydropower plant and the Djamboutou thermal power plant are the plants that supply the peak demand of the RIN. Electricity Production is projected to increase rapidly between 2030 and 2035. This is justified by the fact that the demand to be supplied also increases significantly in the same period. To this extend, the BK_Nord plant gives the total capacity of the plants to be connected to the network to satisfy the demand, i.e. a deficit of 232 MW from 2025.

Fig. 10 shows the evolution of the total capacities installed for the RIN. During the normal operation of all the power plants, the solar power plant is chosen to meet electricity demand. This means that more actions will be needed to increase the rate of construction of

solar power plants in the region. Therefore, to solve this problem, it is suggested that solar power plants must have a total installed capacity of 50 MW as soon as they are commissioned in 2025. The installed power, therefore, increases from 50 MW in 2025 to 125 MW in 2045. To reach the level of generation as shown in the previous figure, it would be necessary to plan for the construction of additional plants depending on the availability of resources in this system.

- Results for the RIS

The evolution of electricity production in RIS is shown in Fig. 11. It can observe the predominance of hydropower plants with installed capacities of more than 200 MW. It is the part of the country with the greatest hydro potential. This part also has three cities whose population growth rate is very high compared to the other cities; Douala, Yaoundé, and Bafoussam. Consequently, the increase in population will lead to an increase in the production of waste. This is what justifies the presence of the two biomass power plants in the energy mix. They must be set at the start of the [2020–2025] period, which will bring 66 MW of power to the

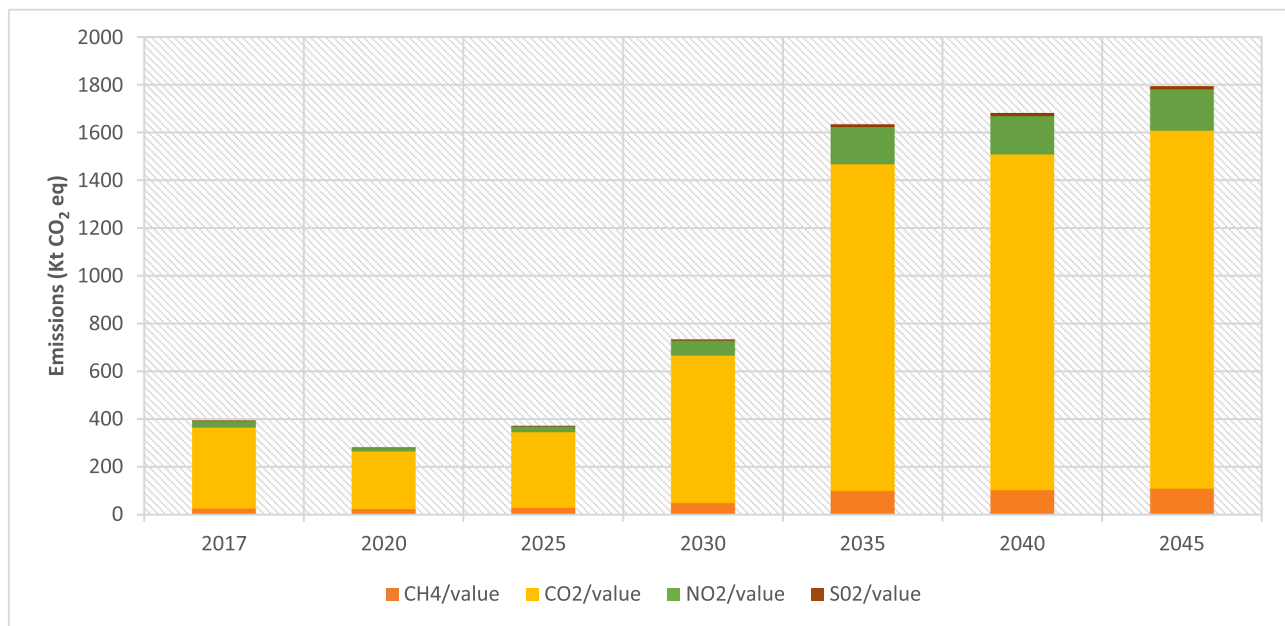


Fig. 18. Emissions of RIE, Baseline Median Scenario.

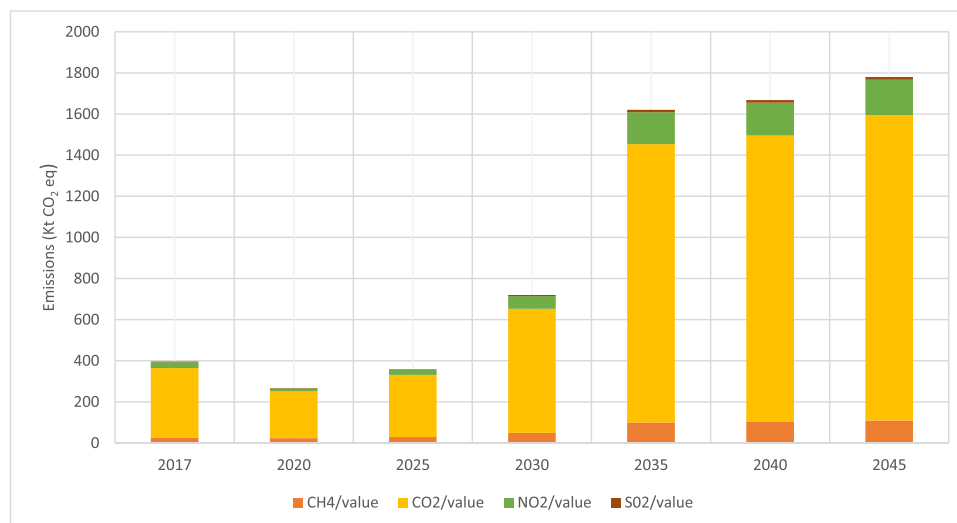


Fig. 19. Emissions of RIE, Sustainable Median Scenario.

network. The direct consequence will be to help satisfy demand but also reduce dependence on fossil fuels. The recovery of biomass will also have the effect of contributing to the sanitation of these different cities.

Despite the commissioning of the planned Nachtigal and Song Mbengue power plants at the start of the period [2025–2030], the country will have an energy deficit in the period from 2035 to 2045. This deficit is materialized by the BK_South power plant. For this request, it would be imperative to commission the Nachtigal and Song Mbengue power plants at their maximum capacity as illustrated in the figure below. This is explained by the fact that, in the event of the completion of the RTA project, the construction of a cement plant in Kribi and plans to extend the park of the autonomous port of Kribi, the current power plants will not be able to meet the new demand from these projects (PDSE2030 update, 2015).

Fig. 12 shows the evolution of the total installed capacities of the RIS. The installed capacities increase over time from 2017 to 2045. The two power plants of the future, namely Song Mbengue and Nachtigal, are those chosen by the model in order to meet demand. This choice is made

by comparing investment and operating costs, the environmental impact, but also by production capacity compared to other power plants. It is clear that the commissioning of the Nachtigal, Song Mbengue, and Grand-Ngodi power plants will allow the supply of energy to large structures in projects.

6.1.2. Median scenario with GHG limitation (Sustainable scenario)

- Results of RIS

Fig. 13 shows the production and installed capacity planning of the RIS for the Sustainable Median scenario. The plants that do not have a contribution significant enough to the energy mix and that produce greenhouse gases, namely the LFO plants of Oyomabang in Yaoundé and Bassa in Douala are to be descommissioned. This plants could be gradually replaced by biomass plants in Douala and Yaoundé. The aim is to develop electricity from the biomass produced in these cities and gradually reduce the dependence on thermal oil power plants. It is therefore required to build biomass power



Fig. 20. GHG emission in various scenarios.

plant with install capacities of 20 MW and 25 MW respectively. More than 36% of the new capacity come from hydroelectricity power plants.

- Results of RIE

For the RIE as for the other networks, the policy is to reduce dependence on fuel oil by exploiting the available biomass. The unavailability of isolated plants during some periods requires an increase in the generation capacity of the RIE biomass plant. Its capacity increases from 12.7 MW for the median baseline scenario to 17.85 MW for the median sustainable scenario Fig. 14. In this figure, biomass contributes about 6% of this generation.

6.2. Results of GHG emissions

- Results of RIS

Fig. 15 shows the evolution of GHG emissions for the RIS for the baseline median scenario and Fig. 16 for the sustainable median scenario. The high emission rate is due to the density of the network. Despite the great hydro potential in this area, which is characterized by a large concentration of hydropower plants, it is also the area with the largest number of thermal oil power plants and gas power plants. These plants are mostly located in localities that are very far from the transformer or basic distribution stations. SO_2 emissions are very low compared to those of other gases and almost do not appear in these figures for most scenarios and networks. Significantly released gases

are CO_2 , CH_4 , and NO_x . The CO_2 is the dominant gas among the three. The increase in the quantities of gas emitted from the 4th period (2030) to the last period (2045) is strongly linked to the constantly growing demand. This increase is observed in all the networks and all the scenarios of this study. There is also a slight decrease in the quantities of GHG emitted in the period (2019–2020), which is explained by the fact that the Mevem'ele plant brings new life to the network, by reducing the activity of certain oil thermal power plants. A considerable decrease in emitted quantities is observed in the first period (2017) for all the scenarios. This is justified by the fact that the already constructed Memvele hydro-power plant will come into operation in this period. Because in this period, the demand is still low, the Memvele power plant could replace the Kribi gas power plant for example.

The definitive replacement of oil thermal power plants of the Oyomabang and Bassa with the biomass power plants of Yaoundé (Biomass_Ynd) and Douala (Biomass_Dla) makes it possible to reduce the trend in emissions from 31046 kt CO_2 eq for the baseline median scenario to 30584 kt CO_2 eq for the sustainable Median scenario in the planning period. Greenhouse gas emissions here are considered zero because these quantities can be absorbed by photosynthesis. A total of 462 kt CO_2 eq can be avoided in the RIS. This calculation was made by considering only the last period which presents high rates of GHG quantities and can extend to other periods.

- Results of RIN

Fig. 18 shows the GHG emissions from the production of electrical

energy in the RIN. The relatively high CO_2 emissions are due to the presence of oil thermal power plants in the electricity mix. This figure also shows a significant increase in the quantities of GHGs. The quantities of CO_2 emitted, for example, go from 1503.4 kt CO_2 eq in 2017–5993.2 kt CO_2 eq in 2045, i.e. 84.75% of the GHG emissions of the same year; increase linked to growth in energy demand. This percentage confirms the thesis that CO_2 emissions are greater than those of other greenhouse gases. This result is validated by the previous study by [Guemene Dountio et al. \(2016\)](#). Mainly in agricultural areas, biomass is used for cooking needs.

- Results of RIE

[Fig. 19](#) and [Fig. 19](#) present the emission of GHGs in RIE for the basic median scenario and sustainable median scenario respectively. As a result, by 2050, total GHG emissions will drop from 1794 kt CO_2 eq for the baseline median scenario ([Fig. 18](#)) to 1780 kt CO_2 eq for the sustainable median scenario ([Fig. 19](#)). This avoids an average of 13.8836 kt CO_2 eq per year at this network, particularly in 2045. [Fig. 20](#) shows a comparison of GHG emissions for the two scenarios.

6.3. Advantages and limits of the study

- The model provides simple management of the energy mix, giving priority to renewable energies;
- About the tool, the value of the objective function of the solver helps to choose the solution considered best according to the criteria specified;
- Produce locally the electricity is to reduce it lost due to transmission,
- Valorization of biomass in electricity generation is to avoid poor waste management practices such as open burning.

However, this study does have its limitations, particularly with regard to fuel supplies. Logging areas are sometimes far from towns and cities. As a result, the plant's operating costs increase due to transport. The same applies to refuse collection areas. If the planned power plants are not built and commissioned, the forecasts will no longer be valid over the planning horizon. In this case, the model will have to be rethought.

7. Conclusion

Renewable energy in Cameroon has great potential to enhance the

electricity supply and reduce GHG emissions. This study considered the three interconnected networks in Cameroon separately to take into account the local situation. Hydroelectricity remains the predominant type of energy in the electricity mix regardless of environmental policy. Two sub-scenarios, the first without limitation of GHG emissions (baseline scenario) and the second with limitation of GHG emissions (sustainable scenario) were developed for each network. Based on these scenarios, a detailed model was developed to estimate the reduction potential for GHG by introducing biomass power plants. The introduction of biomass was also to reduce the transmission of hydroelectricity. The emissions of GHG as the principal constraint were analyzed. Each scenario was built with the emissions of CHG as the principal constraint, maintaining Cameroon at low rates of pollution. Some thermal oil power plants were replaced by thermal biomass power plants in RIS and RIE. Results showed that new energy technologies will play a significant role in Cameroon's future electricity generation, and will have a beneficial effect on GHG emissions in Cameroon. Scenario analysis results also show that in RIS and RIE thermal biomass power plants and hydroelectricity can be instrumental in supplying electricity demand. The key values obtained were 462 kt of CO_2 eq and 14 kt of CO_2 eq. These values represent the amounts of GHGs that can be avoided in the RIS and RIE respectively. We want to extend our study by taking the following actions:

- raising awareness among decision-makers about the need for rational waste management;
- Carry out studies of the availability of biomass for other cities for electricity generation;
- Produce planning models for each of these cities;
- Contribute to the production of reports on the country's GHG emissions.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

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Appendix A

Table A1
Characteristics of existing power plants.

Name	Code	PrimarySource	Efficiency	Live time (Year)	Availability	Power (MW)	Date of commissioning	Var Cost. (US \$/kWyr)	Fixed Cost (US\$/kW/yr)	d'Inv. Cost (US\$/kW)
RIS										
Bassa	EPGas_Bassa	GAZ	0.33	20	0.90	20	2015	28	22	1700
Logbaba	EPGas_Log	GAZ	0.33	20	0.91	30	2015	28	22	1700
Kribi	EPGas_Kribi	GAZ	0.33	25	0.90	216	2013	25,7	19,6	1570
Limbe	EPOHF_Limbe	HFO	0.39	30	0.65	85	2004	20	22	1176.5
Logbaba2	EPOHF_Log	HFO	0.385	30	0.80	12	2001	22	25	1083
Oyombang1	EPOHF_Oyom	HFO	0.38	30	0.50	18	2000	22	25	1108
Dibamba	EPOHF_Dibamba	HFO	0.38	30	0.9	86	2010	20	22	1436
Ahala	EPFlo_Ahala	LFO	0.35	25	0.8	60	2016	15	24	1078
Bafoussam	EPLfo_Baf	LFO	0.35	20	0.75	14.20	2004	16	25	
Bamenda	EPLfo_Bamenda	LFO	0.35	20	0.8	20	2011	16	25	1078
Bassa2	EPOLFEXBassa	LFO	0.35	20	0.75	9	2009	17	28	846
Edéa	EPHYDEXEdéa	Hydro	-	65	0.81	276,2	1975	151	18	1200
Oyombang2	EPOLFEXOyom	LFO	0.34	20	0.80	7.5	2002	17	26	946
Mbalmayo	EPLfo_Mbyo	LFO	0.35	20	0.7	10	2011	17	25	1078
Ebolowa	EPOLFEXEbwa	LFO	0.35	30	0.7	10	2011	17	28	1078

(continued on next page)

Table A1 (continued)

Name	Code	PrimarySource	Efficiency	Live time (Year)	Availability	Power (MW)	Date of commissioning	Var Cost. (US \$/kWyr)	Fixed Cost (US\$/kW/yr)	d'Inv. Cost (US\$/kW)
Mekin	EPHYDNCMe	Hydro	-	60	0.90	15	2017	180	20	2600
Menve'ele	EPHYDNCMenvel	Hydro	-	60	0.90	210	2018	155	20	2300
Songloulou	Hydro_Song	Hydro	-	65	0.98	384.00	1991	135	18	1124
Self product.	SELFPro_South	-	-	-	-	-	-	-	-	-
Beckup	BK_south	-	-	-	-	-	-	-	-	-
Nachtigal	Hydro_Nachtigal	Hydro	-	-	-	-	-	-	-	-
Ngodi	Hydro_Ngodi	-	-	-	-	-	-	-	-	-
Song Mbeng	Hydro_NUSongB	Hydro	-	-	-	-	-	-	-	-
BiomassYde	Biomass_Yde	Biomass	-	-	-	-	-	-	-	-
BiomassDla	Biomass_Dla	Biomass	-	-	-	-	-	-	-	-
RIE										
Bertoua	EPLfo_Bertoua	LFO	0.35	20	0.65	13.6	2013	16	25	953
Isolées	EPLfo_isolees	LFO	0.35	20	0.75	14	2006	17	25	503
Lom Pangar	Hydro_Lompangar	Hydro	-	50	0.95	30	2016	180	20	2500
Biomass	Biomass_East	-	0.36	20	...	15	2025	200	20	
Self product.	SELFPro_East	-	-	-	-	-	-	-	-	-
Beckup	BK_East	-	-	-	-	-	-	-	-	-
RIN										
Lagdo	Hydro_Lagdo	Hydro	-	50	0.93	72	1983	160	19	1150
Djamboutou	EPLfo_Djamboutu	LFO	0.35	25	-	-	-	16	25	-
Beckup	BK_North	-	-	-	-	-	-	-	-	-
Solar	Solar_ProjNorth	solar	-	30	-	35	2021	30	270	4800
Self product.	SELFPro_North	-	-	-	-	12	-	-	-	-

Table A2

Characteristics of Projects of hydroelectric power plants in PDSE 2030.

AménagementHydroélectriques du RIS	Hauteur de chute (en m)	Débit d'équipement (en m ³ /s)	Puissance installée (en MW)	Production en 1 ^{ère} Phase (en GWh)	Production en 2 ^{ème} Phase (en GWh)
SANAGA: NACHTIGAL Amont KIKOT	4450452881422942	7501 4001 4001 4001	250/280350/	2 000/21002800/3	2 2003 8003 5002
Aval NGODI PETIT EWENG SONG		4001 4001 4001 400	550475300450/900384/	500— 3500/	1006 4003 7002
MBENGUESONG LOULOU SONG			530250/300450	50002 9001850-	2003 300
NDONG GRAND EDEA					
MBAM & NOUN: NYANZOM BAYOMEN NOUN-	80786004590	650715230125125	37547012004590	19005 200250500	-2 1002 500-
WOURI BANGATE amont BANGATE aval					
NTEM: MENVE'ELE	56	270/450	120/270	800/1 000	—
NYONG: NJOCK	88	185/370	120/270	750	1400

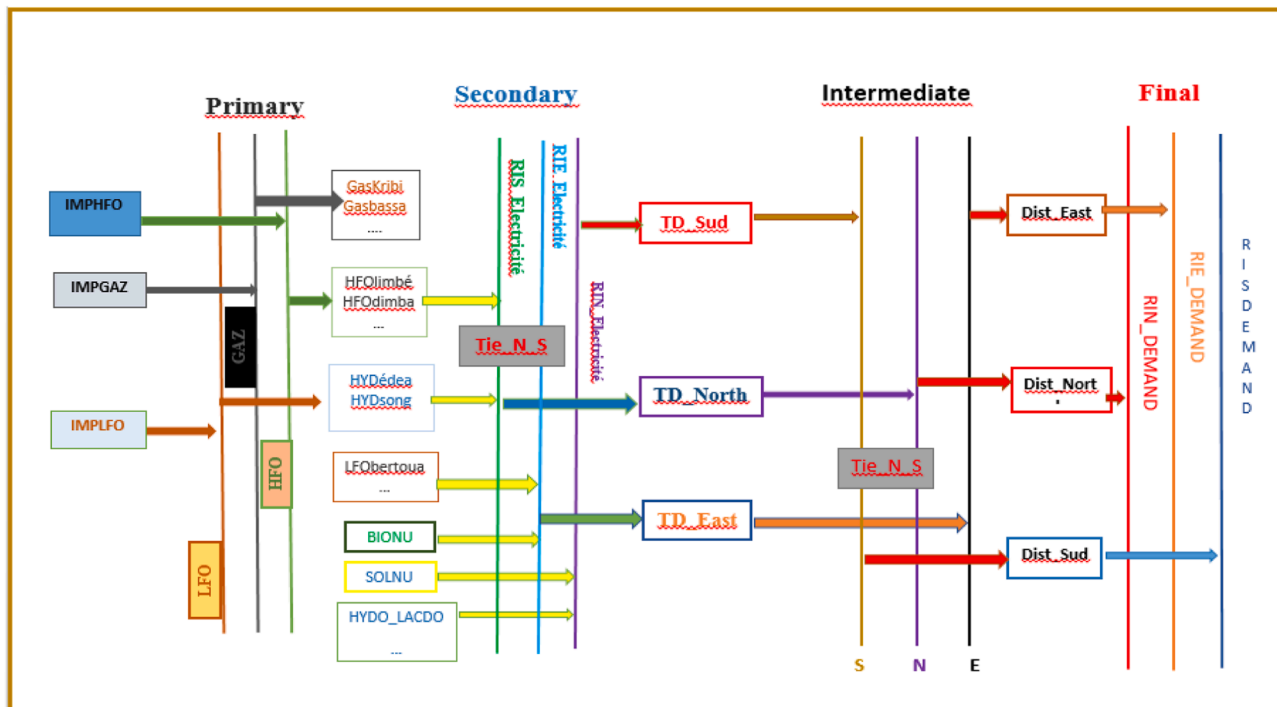


Fig. A1. Energy chain of Cameroon modeled in the study.

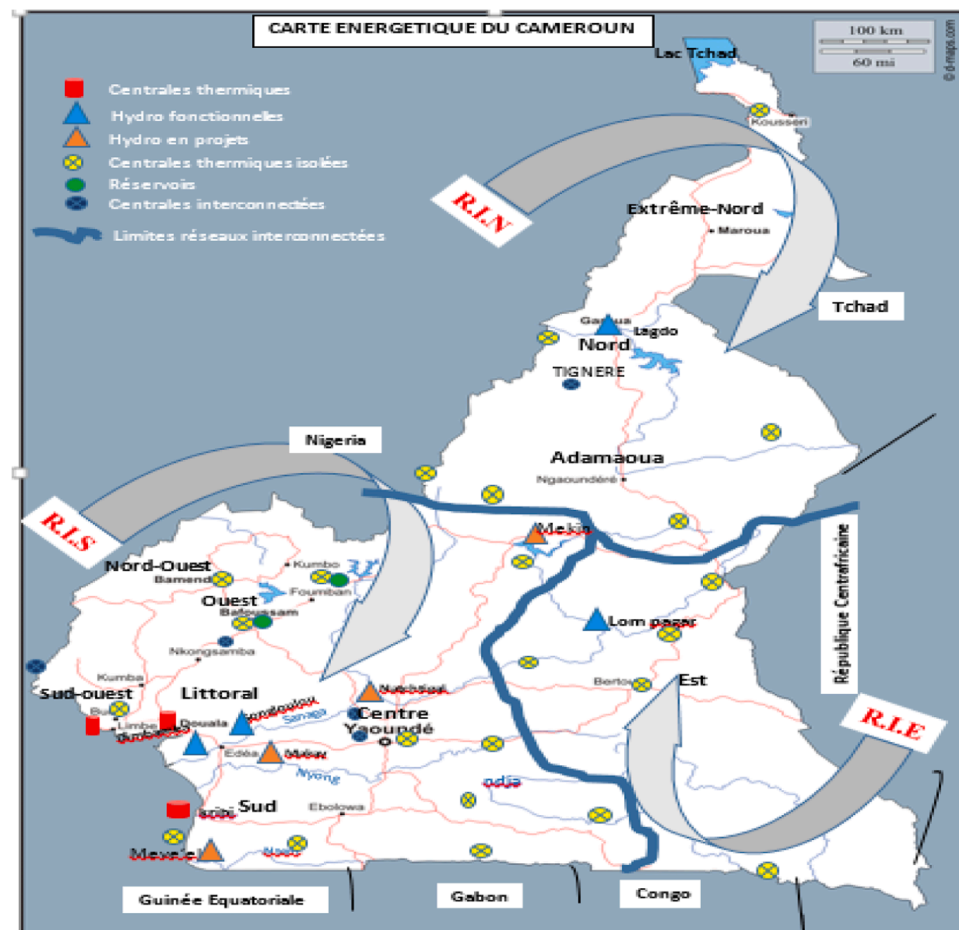


Fig. A2. Energy map in Cameroon.

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